Commodity Prices, Shadow Wages, and Household Labor Allocation Under Imperfect Labor Markets

Shahadat Hossain*

November 3, 2026

Abstract

This paper examines how local poultry prices relate to gendered labor allocation in rural Bangladesh, where limited access to labor markets and gender-specific constraints may influence household production and labor-supply decisions. Using three rounds of the Bangladesh Integrated Household Survey, we estimate gender-specific shadow wages and household shadow income from a poultry production function and examine labor-supply responses using instrumental variables with limited-information maximum likelihood (IV-LIML). Separability tests reject the restrictions implied by separable household decision-making. First-stage estimates show that the relationships between local chicken and egg prices and shadow wages and income vary with household assets and initial poultry capital. Higher female shadow wages are associated with increased poultry participation and working hours, but not with statistically significant changes in overall labor supply. Male shadow-wage coefficients are statistically insignificant, while higher household shadow income is associated with lower overall labor-force participation for both genders. These findings suggest potentially offsetting labor-supply incentives: higher returns to family labor encourage poultry engagement, whereas greater household income may reduce labor-force participation. The concentration of women's responses in poultry production, where they undertake most routine management activities, suggests that women may respond to economic opportunities compatible with household responsibilities and mobility constraints, even without a detectable expansion in overall labor supply.

*Department of Agricultural and Consumer Economics, University of Illinois Urbana-Champaign. Email: sh52@illinois.edu

1 Introduction

Agricultural prices shape rural livelihoods in developing countries by influencing farm profitability, household income, and returns to family labor. Changes in output prices affect production and labor allocation incentives, with implications for agricultural commercialization, market integration, and price-related policies (Bellemare and Wichman, 2020; Lederman and Porto, 2016). For smallholder households, benefiting from market opportunities depends on their ability to adjust production and allocate resources in response to price incentives. These adjustments are particularly relevant where production relies heavily on family labor and production, consumption, and labor-supply decisions are closely linked (de Janvry et al., 1992; Dorward, 2013; Singh et al., 1986). Under imperfect rural labor markets, the value of marginal productivity of family labor, defined as the household-specific shadow wages, may differ from market wages, making labor allocation dependent on household production conditions and agricultural prices (de Janvry et al., 1992; Skoufias, 1994).

Labor-supply responses to agricultural price changes depend on the functioning of rural labor markets. Under complete and competitive markets, household production and consumption decisions are separable, and market wages determine the opportunity cost of family labor (Singh et al., 1986). When labor markets are imperfect, however, these decisions may become nonseparable, and household-specific shadow wages may better reflect the marginal returns to family labor (de Janvry et al., 1992; Dillon et al., 2019; Skoufias, 1994). In this setting, higher output prices may increase shadow wages and encourage greater labor allocation to household production, while increased household income may reduce labor supply through an income effect. The net response depends on the relative strength of these opposing effects, which may vary with household production conditions and labor-market constraints. Thus, the direction and magnitude of labor-supply responses to agricultural price changes remain empirical questions.

Smallholder poultry production in rural Bangladesh provides a relevant setting for examining household responses to expanding agricultural markets. Despite increasing commercialization of the poultry sector (Hennessey et al., 2026), many rural households continue to raise small flocks of indigenous or crossbred birds for consumption and sale (Chowdhury, 2013; FAO, 2004). Poultry production requires relatively little land and can be undertaken near the homestead, making family labor a central production input. Labor responsibilities are strongly differentiated by gender, with women primarily managing routine poultry activities at the homestead and men more involved in input purchases and marketing (Nath et al., 2025; Rimi et al., 2016). These patterns are reinforced by social norms and mobility constraints that limit women's access to employment outside the household (Heintz et al., 2018). Although expanding poultry markets create potential income opportunities, increasing production may require additional investments in housing, feed, veterinary inputs, and technical skills, potentially constraining smallholders' ability to respond to price incentives. Thus, household labor responses to poultry-market opportunities may depend on both productive resources and gender-specific constraints.

Under this context, an important question is whether the expansion of poultry markets has created economic incentives that are reflected in smallholder households' production and labor allocation decisions. If poultry production represents a meaningful market-oriented activity, variation in local output prices may be associated

with changes in the implicit returns to family labor and the allocation of household working time. However, the extent of these adjustments may depend on household production capacity, available resources, and gender-specific constraints. This study examines whether changes in local chicken and egg prices influence household-specific shadow wages, shadow income, and the allocation of female and male family labor in rural Bangladesh. Drawing on the nonseparable agricultural household framework, changes in poultry prices may affect gender-specific shadow wages and household shadow income through their effects on poultry production returns. Changes in shadow wages and shadow income may, in turn, influence labor allocation through substitution and income effects, making the net labor-supply response theoretically ambiguous. These responses may differ between women and men because of differences in productive responsibilities and alternative employment opportunities. The analysis therefore examines both poultry-specific and overall labor force participation and working hours to assess how household labor allocation responds to changes in the economic returns to poultry production.

A substantial literature examines how agricultural prices influence household production, labor allocation, and welfare in developing countries (Beck et al., 2019; Benjamin, 1992; Danzer and Grundke, 2020; Miller and Urdinola, 2010). While these studies provide evidence on household responses to changing market conditions, observed labor adjustments may reflect changes in both the returns to family labor and household resources. Distinguishing these mechanisms is important because higher returns to family labor may encourage greater engagement in household production, while increased household resources may reduce labor supply through an income effect. Understanding their relative roles can help explain why households respond differently to similar price incentives and whether expanding agricultural markets lead to greater labor participation or shifts in labor allocation across activities. However, identifying these relationships is challenging under imperfect labor markets, where production and labor-supply decisions are jointly determined and the extent of nonseparability varies across agricultural settings (Dillon et al., 2019).

The agricultural household literature examines how nonseparability links production and labor allocation decisions, with household-specific shadow wages potentially differing from market wages (de Janvry et al., 1992; Rosenzweig, 1980; Singh et al., 1986). Using this framework, Jacoby (1993) estimates shadow wages from the marginal value product of family labor and examines their relationship with labor supply among agricultural households in the Peruvian Sierra. Similarly, Skoufias (1994) estimates gender-specific shadow wages and shadow income to analyze labor supply in rural India. Subsequent studies extend these approaches to settings with imperfect rural factor markets (Abdulai and Regmi, 2000; Barrett et al., 2008; Makhdum and Kousar, 2021). While this literature establishes the importance of household-specific opportunity costs, much of the evidence comes from crop-based agricultural settings, with limited attention to how local poultry prices relate to gender-specific shadow returns and labor allocation in smallholder poultry production. Addressing this gap helps assess whether expanding poultry markets influence household labor decisions and how these relationships vary with productive resources and gender-specific employment constraints.

This paper connects the agricultural-price and shadow-wage literatures by examining how local poultry prices relate to household-specific shadow wages, shadow income, and gender-specific labor allocation. As poultry production in rural Bangladesh becomes increasingly market-oriented (Hennessey et al., 2026), an important

question is whether local price incentives are reflected in household labor decisions, indicating the extent to which smallholders engage in poultry production as an economic activity. These relationships may operate through changes in the marginal value product of family labor and household resources, with potentially different implications for labor supply. Moreover, households with different asset holdings and initial poultry capital may respond differently to similar price movements. Given women's substantial involvement in household poultry production, these adjustments may also differ by gender and between poultry-specific and overall labor supply. Examining these relationships provides insight into whether expanding poultry markets create meaningful economic opportunities for smallholder households and how these opportunities may be reflected in their labor allocation decisions.

The analysis uses three rounds of the Bangladesh Integrated Household Survey (BIHS), conducted in 2011/12, 2015, and 2018/19, within a nonseparable agricultural household framework. A poultry production function is estimated using female and male family labor, initial poultry capital, household productive assets, and other production inputs. Following Jacoby (1993) and Skoufias (1994), gender-specific shadow wages are derived from the marginal value products of family labor, while household shadow income is constructed from predicted poultry returns and non-poultry income. These measures are incorporated into gender-specific labor-supply equations examining poultry participation and hours, overall labor-force participation, and total working hours. To address potential endogeneity, the labor-supply equations are estimated using instrumental variables with limited-information maximum likelihood (IV-LIML), exploiting variation in local chicken and egg prices, household assets, and their interactions. This approach allows the relationships between shadow wages, shadow income, and labor allocation to be examined separately, subject to instrument relevance and validity.

The empirical analysis yields three main findings. First, the estimated production function highlights the importance of family labor in smallholder poultry production, while separability tests reject the restrictions implied by separable household production and consumption decisions, supporting the relevance of household-specific shadow wages. Second, the first-stage estimates indicate that the relationships between local poultry prices and shadow wages and income vary with household assets and initial poultry capital, suggesting that similar price movements may generate different implicit returns across households. Third, the IV-LIML estimates reveal gender-specific patterns in labor allocation. Higher female shadow wages are positively and significantly associated with poultry participation and working hours, but show no statistically significant relationship with overall labor-force participation or total working hours. This pattern indicates a possible reallocation of women's labor toward poultry production, although changes in non-poultry labor are not directly established. Male shadow wages, by contrast, show no statistically significant associations with poultry-specific or overall labor outcomes, while higher shadow income is associated with lower overall labor-force participation for both women and men. Together, these findings suggest that expanding poultry markets may create opportunities for greater female engagement in household production without necessarily generating a corresponding increase in overall labor supply, highlighting the potential importance of household production capacity and gender-specific constraints in shaping labor responses to market incentives.

The paper makes three contributions to the literature. First, it extends the literature on nonseparable agricultural

households and shadow wages by examining how local output prices and household productive endowments relate to gender-specific shadow wages and household shadow income (Abdulai and Regmi, 2000; Barrett et al., 2008; Jacoby, 1993; Skoufias, 1994). While existing studies establish the importance of household-specific opportunity costs for labor allocation, less attention has been given to how local agricultural prices are reflected in these implicit returns. By linking chicken and egg prices to shadow wages and income, the analysis provides evidence on the relationship between output-market conditions and household labor incentives under imperfect rural labor markets.

Second, the paper contributes to understanding the economic role of smallholder poultry production in rural Bangladesh. Previous studies document the contributions of backyard poultry to household income, consumption, and livelihoods (Islam and Jabbar, 2005; Nielsen et al., 2003; Shanta et al., 2017), while recent evidence highlights the increasing commercialization of the poultry sector (Hennessey et al., 2026). However, less is known about how local poultry-market incentives relate to labor allocation within smallholder households. The findings indicate that higher female shadow wages are associated with greater poultry participation and working hours, without a statistically significant corresponding relationship with overall labor supply. These results provide evidence of poultry's role in household economic activity and suggest that greater engagement in poultry production need not be accompanied by a detectable expansion in total working hours.

Third, the paper contributes to the literature on gendered labor allocation in smallholder agriculture and livestock production. Existing studies document gender differences in agricultural responsibilities, access to productive resources, and employment opportunities (Doss, 2017; Doss et al., 2018; Palacios-Lopez et al., 2017), alongside women's substantial involvement in livestock production (Swaminathan and Usami, 2016; Vijayamba and Swaminathan, 2024). In rural Bangladesh, where social norms and mobility constraints shape women's employment opportunities (Heintz et al., 2018), family-based poultry production provides a relevant setting for examining gender-specific labor responses to market incentives. The findings show that female shadow wages are positively associated with poultry participation and working hours, whereas comparable statistically significant relationships are not detected for men. These differences highlight the importance of accounting for gender-specific production incentives and labor constraints when examining household responses to agricultural market opportunities.

The findings have implications for agricultural commercialization, livestock value-chain development, and policies that influence producer incentives. Changes in local poultry-market conditions may affect household labor allocation through both the implicit returns to family labor and household resources, with the resulting adjustments depending on productive endowments and existing gender divisions of labor. Policies that improve returns to poultry production may therefore be associated with greater engagement in household poultry activities, particularly among women, without necessarily increasing overall labor supply. At the same time, changes in household resources may influence participation in income-generating work in the opposite direction. Evaluations of livestock-market interventions should consequently consider gender-specific labor allocation and household production behavior alongside changes in output, income, and market employment. Because additional poultry work may also involve trade-offs with other productive and unpaid activities,

increased participation should not automatically be interpreted as an improvement in women's welfare.

The paper proceeds as follows. Section 2 describes Bangladesh's poultry sector, Section 3 presents the conceptual framework, Section 4 describes the data and construction of local poultry prices, Section 5 outlines the empirical strategy, Section 6 presents the results and robustness checks, and Section 7 concludes.

2 Institutional Background and Poultry Sector Context

2.1 Growth and Commercialization of the Poultry Sector

Bangladesh's poultry sector has expanded rapidly over the past three decades, driven by rising incomes, population growth, urbanization, and increasing demand for animal-source foods (Ecker and Comstock, 2021). Chicken and eggs provide relatively affordable sources of animal protein, while poultry production offers shorter production cycles and lower entry requirements than larger livestock. Reflecting this expansion, national egg production increased fourfold, from approximately 6.1 billion eggs in 2010–11 to 24.4 billion in 2024–25 (DLS, 2025).

This growth has been accompanied by increasing commercialization, with the expansion of commercial broiler and layer farms, hatcheries, feed mills, veterinary services, and marketing networks. Nevertheless, household-based poultry production remains widespread. Many rural households continue to raise indigenous and crossbred chickens, while the expansion of Sonali production, based on a cross between Rhode Island Red and Fayoumi breeds, illustrates the coexistence of commercializing poultry markets and decentralized production systems linked to smallholders (Hennessey et al., 2026). Bangladesh produces approximately 140 million Sonali and 500 million broiler chickens annually (Hennessey et al., 2026). According to the 2017–18 Bangladesh Demographic and Health Survey (BDHS), 68.4% of rural households owned chickens or ducks (National Institute of Population Research and Training (NIPORT) and ICF, 2020), highlighting the continued prevalence of household poultry ownership.

The coexistence of commercial and household-based production raises an important question about the extent to which expanding poultry markets create economic opportunities for smallholders. In particular, whether local poultry prices are reflected in household-specific production returns and labor allocation depends partly on households' productive resources and their capacity to adjust production. These relationships motivate the paper's focus on household poultry production and gender-specific labor allocation in rural Bangladesh.

2.2 Household Poultry Production and the Gender Division of Labor

Smallholder poultry production in rural Bangladesh is typically organized around small, multipurpose flocks that provide eggs, meat, and occasional cash income. Unlike specialized commercial broiler and layer operations, backyard systems commonly combine breeding, egg production, and raising birds for consumption or sale within the same household enterprise. In a large rural Bangladesh study, households kept a median of seven chickens, with an interquartile range of three to twelve, and most birds scavenged for food during the day (Shanta et al., 2017). Hens may be retained for egg production and breeding, while other birds are raised for sale or household consumption. Thus, chicken and egg production are linked through common flock resources, family labor, and decisions about retaining, consuming, or selling birds and eggs.

Production follows a biological cycle that shapes the timing of household decisions. Indigenous hens generally begin laying at approximately six to seven months of age and produce eggs in successive clutches rather than continuously. Evidence from Bangladesh indicates that indigenous hens commonly produce around three clutches and approximately 46 eggs annually, with a production cycle lasting roughly 105–140 days (Sarkar and Golam, 2009). Eggs retained for natural incubation hatch after approximately 21 days, followed by a period of chick rearing during which the hen's egg production may be interrupted. Consequently, decisions to retain eggs for hatching, raise chicks, or sell mature birds affect both current output and future flock productivity. Production may also vary seasonally with feed availability, temperature, disease exposure, and chick survival. Evidence from the Sylhet region suggests that egg production is relatively higher during winter, although seasonal patterns may differ across locations and production systems (Khan et al., 2006).

Households can adjust poultry production by purchasing or retaining additional birds, providing supplementary feed, improving housing and veterinary care, and changing the allocation of family labor. These adjustments differ in timing and resource requirements. Additional feeding and care may affect bird health, survival, and productivity within an existing flock, whereas expanding the number of productive hens requires either purchasing mature birds or allowing chicks to reach reproductive age. Such decisions also involve trade-offs between current sales and future production, particularly when hens are retained for laying rather than sold for meat. Accordingly, poultry labor requirements may vary with flock size, flock composition, and the intensity of management, although the relationship need not be proportional because several routine tasks can be performed jointly.

Family labor is central to this production system and is strongly differentiated by gender. Women typically undertake routine homestead-based activities, including feeding, watering, cleaning shelters, collecting eggs, and caring for chicks, while men are more involved in purchasing inputs, transporting birds, and marketing poultry products (Nath et al., 2025; Rimi et al., 2016). These differences reflect both production responsibilities and gender-specific opportunities outside the household. Social norms and mobility constraints can limit women's access to wage and nonfarm employment (Heintz et al., 2018), making home-based poultry activities a potentially important source of productive work. An increase in flock size or management intensity may therefore raise women's routine labor requirements, while changes in purchases and sales may affect men's market-related activities.

These production characteristics motivate the empirical framework. Chicken and eggs are treated as outputs of a common household poultry production technology because both depend on shared flock resources and labor inputs, even though their production and sale occur at different points in the biological cycle. Local chicken and egg prices may influence the value of poultry output, the returns to retaining or selling birds, and the marginal value product of family labor. Their effects on household labor allocation may nevertheless differ because production adjustments require time, productive resources, and gender-specific activities. The analysis therefore considers both output prices and estimates female and male shadow wages separately, allowing the relationships between poultry-market conditions and household labor allocation to be examined within a nonseparable agricultural household framework.

2.3 Local Poultry Markets and Price Variation

Poultry products in Bangladesh are marketed through decentralized trading networks that connect household producers and commercial farms with local collectors, intermediaries, wholesalers, retailers, and consumers (Islam et al., 2017; Larive International B.V. and LightCastle Partners Ltd., 2023). Smallholders typically sell birds and eggs at the farmgate or in nearby village and subdistrict markets, and traders link these markets to larger regional trading centers. Marketing channels differ across production systems: indigenous and backyard chickens often pass through different channels than commercial broilers and layers (Islam et al., 2017; Moyen et al., 2018). Despite improvements in transport and connectivity, differences in transport costs, access to traders, and marketing arrangements generate spatial differences in the prices producers receive. Local poultry prices therefore reflect both broader market conditions and the characteristics of individual trading networks.

Local poultry prices fluctuate with seasonal demand and supply conditions. Religious festivals, particularly Ramadan and Eid, can increase demand for poultry products and generate temporary price movements (Islam et al., 2017; Sattar et al., 2021). On the supply side, changes in feed costs and disease outbreaks affect production costs, flock availability, and market prices (Høg et al., 2021). Backyard poultry production is also subject to seasonal variation in feed availability, disease exposure, and reproductive cycles, although households' ability to expand production in response to temporary price increases may be limited by flock size and biological production cycles (Khan et al., 2006; Sarkar and Golam, 2009). These conditions may affect local markets differently depending on production intensity, market access, and trading networks. In the empirical analysis, local chicken and egg prices are measured at the beginning of the annual production period, preceding the production and labor decisions examined in the study. While this timing reduces concerns about contemporaneous reverse causality, anticipated price changes and seasonal production conditions may still influence household labor allocation.

Market accessibility is a further source of local price variation. Households near major trading centers generally have better access to buyers and input suppliers, whereas producers in remote villages face higher transport costs and fewer marketing options. These differences affect farmgate prices and determine how fully changes in regional conditions reach individual villages. Chicken and egg prices also need not respond identically to shocks, because the two products differ in production cycles, storage and transport requirements, and demand patterns. Live birds bear higher transport costs because they are shipped live and lose weight in transit, whereas eggs can be graded, packed, and transported more efficiently, supporting wider marketing networks and more spatial arbitrage (Larive International B.V. and LightCastle Partners Ltd., 2023; Sonaiya and Swan, 2004). Households may therefore face different combinations of chicken and egg prices across locations and periods. This variation changes the return to household poultry production and, through it, the value of family labor, which is the margin examined in the rest of the paper. Section 4 describes how I construct village-level prices, and Section 5 discusses the empirical strategy.

3 Conceptual framework

This study draws on the nonseparable agricultural household model to examine how local poultry prices relate to the allocation of female and male family labor in rural Bangladesh. Agricultural households simultaneously act as producers and consumers, allocating productive resources and time across household production, other economic activities, and leisure (Benjamin, 1992; Singh et al., 1986). Under complete and competitive markets, production and consumption decisions are separable: households maximize agricultural profits at prevailing market prices and subsequently allocate income and time according to their preferences. When labor markets are imperfect, however, the opportunity cost of family labor may differ from observed market wages, making production and consumption decisions interdependent. Household-specific shadow wages therefore become relevant to labor allocation decisions (Jacoby, 1993; Skoufias, 1994).

This framework is particularly relevant to smallholder poultry production in rural Bangladesh, where households rely heavily on family labor and use common productive resources to produce chickens and eggs. As illustrated in Figure 1, local poultry prices are linked to household labor allocation through two related mechanisms. First, changes in output prices alter the value of poultry production and the marginal value product of female and male family labor, potentially changing their household-specific shadow wages. Higher shadow wages may encourage greater labor allocation to poultry production, depending on available employment opportunities and competing household responsibilities. Second, changes in poultry returns may affect household shadow income, potentially reducing labor supply through an income effect. The relative importance of these mechanisms depends on household productive resources, including asset holdings and initial poultry capital, as well as gender-specific labor constraints. Consequently, changes in poultry-market incentives may be associated with different responses in poultry-specific and overall labor supply for women and men.

3.1 Household Poultry Production

Consider a household h that produces chickens and eggs using female and male family labor, poultry capital, and other productive inputs. Let the household's poultry production technology be represented by

$$Q_h = F(L_h^f, L_h^m, K_h, X_h; Z_h), \tag{1}$$

where Q_h denotes aggregate poultry output, L_h^f and L_h^m represent female and male family labor allocated to poultry production, K_h denotes poultry capital, X_h represents other variable inputs, and Z_h captures household-specific production characteristics and environmental conditions.

Household poultry production often involves multipurpose flocks that generate both meat and eggs using shared resources. Hens produce eggs during their productive lives, give birth to new chicks and may subsequently be sold or consumed, while young birds may be retained as replacement stock, raised for sale, or consumed within the household. Consequently, chicken and egg production are technologically interconnected rather than necessarily independent activities. The aggregate production function in Equation (1) provides a representation of this common production technology.

Let P_h^c and P_h^e denote the local prices of chickens and eggs, respectively, and let Q_h^c and Q_h^e denote the corresponding outputs. The total value of household poultry production is $V_h = P_h^c Q_h^c + P_h^e Q_h^e$. The household's net poultry returns can be expressed as $\Pi_h = V_h - E_h(X_h, K_h)$, where $E_h(\cdot)$ denotes monetary production costs, excluding the imputed value of family labor to calculate the net profit from poultry production. Thus, changes in local chicken and egg prices alter the value of poultry production and may influence decisions concerning flock size, input use, and the allocation of family labor. Because the two outputs share productive resources, a change in the price of either product may affect the profitability of the household's overall poultry enterprise.

3.2 Household Optimization and Nonseparability

Households derive utility from consumption and leisure while allocating their available time between poultry production, other productive activities, and nonmarket uses. Let household utility be represented by

$$U_h = U(C_h, R_h^f, R_h^m; \Omega_h), \tag{2}$$

where C_h represents household consumption, R_h^f and R_h^m denote female and male leisure, and Ω_h captures household preferences and demographic characteristics consisting of individual and household characteristics. For each gender $g \in \{\text{female}(f), \text{male}(m)\}$, the time constraint is $T_h^g = L_{h,p}^g + L_{h,o}^g + L_{h,mk}^g + R_h^g$, where T_h^g denotes available time, $L_{h,p}^g$ represents labor allocated to poultry production, $L_{h,o}^g$ denotes labor allocated to other productive activities, and $L_{h,mk}^g$ denotes agricultural and nonagricultural market employment. Nonmarket household responsibilities may also constrain the time available for productive work and leisure.

To derive the shadow wage under imperfect labor markets, consider the benchmark in which family labor cannot be freely hired or supplied at an exogenous market wage. Other productive activities are represented by a net earnings function $B_h(L_{h,o}^f, L_{h,o}^m)$, which may capture agricultural and nonagricultural employment opportunities. The household's consumption constraint is $C_h = \Pi_h + B_h(L_{h,o}^f, L_{h,o}^m) + Y_h^{NL}$, where Y_h^{NL} denotes exogenous nonlabor income.

The household optimization problem is

$$\begin{aligned} \max_{C_h, R_h^f, R_h^m, L_{h,p}^f, L_{h,p}^m, L_{h,o}^f, L_{h,o}^m, X_h} \quad & U(C_h, R_h^f, R_h^m; \Omega_h) \\ \text{subject to} \quad & C_h = \Pi_h + B_h + Y_h^{NL}, \\ & T_h^g = L_{h,p}^g + L_{h,o}^g + L_{h,mk}^g + R_h^g, \qquad g \in \{f, m\}, \\ & (Q_h^{ce}) \in \mathcal{F}(L_{h,p}^f, L_{h,p}^m, K_h, X_h; Z_h). \end{aligned} \tag{3}$$

For the first-order conditions, suppose that the feasible production frontier can locally be represented by differentiable joint-output functions $Q_h^{ce}(\cdot)$, and that the household has an interior optimum. The Lagrangian

is

$$\begin{aligned}\mathcal{L}_h = U(C_h, R_h^f, R_h^m; \Omega_h) \\ + \lambda_h \left[\Pi_h + B_h + Y_h^{NL} - C_h\right] \\ + \sum_{g \in \{f,m\}} \mu_h^g \left[L_{h,mk}^g\right],\end{aligned} \tag{4}$$

where λ_h is the marginal utility of income and μ_h^g is the inequality constraints on the market work for gender g i.e., $L_{h,mk}^g \geq 0$.

The first-order condition with respect to consumption is

$$\frac{U_{R^g}}{U_C} = w_h^{g*} = w_h^g + \frac{\mu_h^g}{\lambda_h} \tag{5a}$$

$$w_h^{g*} = w_h^g \qquad \text{if } L_{h,mk}^g < M^g \tag{5b}$$

$$w_h^{g*} < w_h^g \qquad \text{if } L_{h,mk}^g = M^g \tag{5c}$$

Under nonseparability, labor is allocated in such a way that the marginal product of labor is equal to an endogenously determined shadow wage (w_h^{g*}) for $g \in \{f, m\}$. Thus, we introduce market failure in the model as a market labor constraint: $L_{h,mk}^g \leq M^g$, where M is the maximum number of hours a member can work in the labor market.This constraint is particularly relevant in rural Bangladesh, where limited employment opportunities, labor market frictions, and gender-specific mobility and social restrictions may prevent household members from supplying their desired amount of labor at prevailing market wages. When the constraint binds, the shadow wage falls below the market wage, reflecting the lower marginal value of additional time within the household relative to the wage available in the constrained labor market. Consequently, household-specific shadow wages may differ between women and men, influencing their allocation of labor between poultry production and other economic activities. For example, women may devote more time to routine flock management, feeding, and egg collection, while men may contribute to input procurement and other production-related activities. Differences in productive responsibilities, access to complementary inputs, and household labor constraints can generate gender-specific marginal returns even within the same poultry enterprise.

In the agricultural household framework, using the shadow wage, the household shadow income represents the resources available to the household after accounting for production returns and the implicit value of family labor. Conceptually, it can be expressed as

$$Y_h^* = \Pi_h + B_h + Y_h^{NL} - w_h^{f*} L_{h,p}^f - w_h^{m*} L_{h,p}^m + w_h^{f*} T_h^f + w_h^{m*} T_h^m, \tag{6}$$

where Y_h^{NL} denotes exogenous nonlabor income. This expression is a full-income representation that values available time at household-specific shadow wages. Its exact empirical counterpart depends on how household production returns, labor endowments, and other income sources are measured.

Given shadow wages and shadow income, the structural demand for leisure is expressed as

$$R_h^g = R^g \left(w_h^{f*}, w_h^{m*}, Y_h^*, \Omega_h \right), \qquad g \in f, m, \tag{7}$$

where Ω_h captures household demographic characteristics and preferences. Using the time constraint, the corresponding structural labor-supply function is

$$L_h^g = T_h^g - R^g \left(w_h^{f*}, w_h^{m*}, Y_h^*, \Omega_h \right). \tag{8}$$

Changes in local chicken and egg prices may influence household labor allocation through two interrelated channels: gender-specific shadow wages and household shadow income. The shadow wage channel captures changes in the marginal returns to family labor, while the shadow income channel reflects changes in household resources. Higher poultry prices may increase both, depending on production responses and input costs. An increase in the shadow wage raises the opportunity cost of leisure, potentially encouraging greater labor supply through the substitution effect, while an increase in shadow income may reduce labor supply if leisure is a normal good. Because these effects may reinforce or offset one another, the overall labor response is theoretically ambiguous.

These responses may also differ by gender due to differences in productive responsibilities, domestic time commitments, and access to alternative employment. In rural Bangladesh, women are often engaged in routine family-poultry management, whereas men may combine poultry-related activities with other agricultural and nonagricultural work. Consequently, higher returns to poultry production may encourage women to reallocate time toward poultry activities without necessarily increasing their overall working hours. Male labor responses may differ depending on the relative returns to poultry production and alternative employment opportunities.

To capture these adjustments, I distinguish between poultry-specific and overall labor supply for women and men. While overall labor supply follows from the household's time allocation decision, poultry-specific labor allocation can be expressed as

$$L_{h,p}^g = \mathcal{L}_p^g \left(w_h^{f*}, w_h^{m*}, Y_h^*, \Omega_h \right), \qquad g \in \{f, m\}, \tag{9}$$

where $L_{h,p}^g$ denotes labor allocated to poultry production by household members of gender g. This relationship allows poultry labor allocation to depend on both female and male shadow wages and household shadow income, reflecting the interdependence of labor decisions within the household. Accordingly, the empirical analysis examines poultry-specific and overall labor participation and working hours separately for women and men.

The conceptual framework yields three empirical implications. First, changes in local chicken and egg prices may affect gender-specific shadow wages and household shadow income through their effects on poultry production net-profit. Second, changes in shadow wages and shadow income may influence labor supply through substitution and income effects, with theoretically ambiguous net responses. Third, these adjustments

may differ between women and men and between poultry-specific and overall labor supply, reflecting differences in productive responsibilities and alternative employment opportunities. These implications motivate the empirical analysis of how poultry prices relate to household shadow wages, shadow income, and gender-specific labor participation and working hours.

4 Data, sample construction, and price measurement

4.1 Data

The empirical analysis uses three waves of the Bangladesh Integrated Household Survey (BIHS), a nationally representative panel survey of rural households conducted by the International Food Policy Research Institute (IFPRI) in 2011–12, 2015, and 2018–19 (Ahmed, 2013, 2016, 2020). The BIHS provides detailed information on agricultural production, livestock ownership, production inputs and outputs, individual labor activities, household consumption, and income sources. These data allow us to examine household poultry production and gender-specific labor allocation within a nonseparable agricultural household framework. As illustrated in Figure 2, local chicken and egg prices are measured at the beginning of each annual poultry production period, while production and input-use variables refer to the corresponding 12-month period. In contrast, individual employment outcomes generally refer to the seven days preceding the household interview, conducted after the annual production reference period.

The BIHS employs a two-stage stratified sampling design covering rural Bangladesh. The baseline survey interviewed 5,503 households across 275 villages, with subsequent rounds re-interviewing original households and tracking split households. Panel attrition between the first and third rounds was below 7 percent. The geographic distribution of the surveyed villages is presented in Figure A.1.

The combination of production, labor, and household economic data across survey rounds enables us to estimate poultry production functions, construct gender-specific shadow wages and household shadow income, and examine their relationships with labor allocation. The panel structure also provides variation in local poultry prices and household production decisions over time, supporting an analysis of how changes in poultry prices relate to poultry-specific and overall labor participation and working hours among women and men. The following subsections describe the estimation sample and the construction of village-level chicken and egg prices.

4.2 Estimation Sample

The primary estimation sample consists of a balanced panel of 1,464 poultry-producing households observed in all three waves of the Bangladesh Integrated Household Survey (BIHS). A household is classified as a poultry producer in a given round if it reports producing chickens, eggs, or both. To ensure consistent measurement of poultry production and its associated labor returns, we retain household-round observations with nonmissing initial chicken and egg prices, positive family labor inputs, and total poultry production below the 99.5th percentile of the production distribution. These restrictions yield a balanced panel of households satisfying the eligibility criteria in all three survey rounds.

Household production functions are estimated using information on poultry output, female and male family

labor, production expenditures, and productive assets. The estimated production parameters are used to recover gender-specific shadow wages and household shadow income, which are subsequently linked to individual labor outcomes for household members aged 15–64 years. Labor-supply equations are estimated separately for women and men, distinguishing between poultry-specific and overall labor participation and working hours.

Figure A.2 illustrates the sample selection process, beginning with the full BIHS household sample and identifying eligible poultry producers in each round. Households satisfying the eligibility criteria in at least two rounds form a broader panel used for robustness checks, while those satisfying the criteria in all three rounds constitute the primary estimation sample of 1,464 households (4,392 household-round observations). This approach maintains a consistent sample of poultry-producing households across survey rounds while allowing the robustness of the results to be assessed using a less restrictive panel.

Table A.1 compares the balanced producer panel with poultry-producing households observed in only two survey rounds. The two groups are generally comparable across demographic, socioeconomic, production, and labor characteristics, although some differences are evident in landholdings. In particular, households in the balanced panel own more agricultural land in Round 3, the normalized difference indicates a meaningful difference in this characteristic. While normalized differences for most other variables are relatively small, the agricultural land imbalance suggests that restricting the analysis to the balanced panel may introduce some selection in terms of land ownership. Overall, the comparison indicates broad similarity between the groups, while highlighting potential differences in land ownership.

4.3 Local Poultry Prices

Local chicken and egg prices are constructed from household-reported unit values associated with poultry inventories at the beginning of each annual production period. Specifically, the BIHS records information on poultry inventories at the start of the 12-month production reference period, allowing initial unit values to be distinguished from prices observed during or at the end of production. Household-reported initial unit values are aggregated at the village level to construct local chicken and egg prices, capturing spatial and temporal variation in market conditions facing poultry-producing households. As illustrated in Figure 2, these initial prices precede the annual production period and the subsequent measurement of individual employment outcomes. This temporal ordering reduces concerns about contemporaneous reverse causality between household production decisions and local prices. However, initial prices may still reflect anticipated demand, seasonal market conditions, or unobserved local supply shocks that also influence household production and labor allocation. Consequently, the use of beginning-of-period prices does not fully eliminate potential endogeneity, and their interpretation as instruments relies on the exclusion restriction discussed in the empirical strategy.

Following the literature on recovering spatial price variation from geographically clustered household-reported unit values (Deaton, 1988, 1990), household-specific local chicken and egg prices are constructed using a leave-one-out median of initial unit values reported by other households in the same village and survey round. This approach reduces the influence of household-specific valuations and extreme observations. The local

price assigned to household h for product k is defined as

$$P^k_{hvt} = \text{median}\left\{p^k_{jvt,0} : j \in \mathcal{H}_{vt},\ j \neq h,\ p^k_{jvt,0} \text{ observed}\right\}, \qquad k \in \{c, e\}, \tag{10}$$

where $p^k_{jvt,0}$ is the initial unit value of product k reported by household j in village v and round t, and $\mathcal{H}_{vt}$ denotes the surveyed households in that village-round. The median requires at least three valid observations from other households; otherwise, the price is treated as missing.

Excluding a household's own valuation reduces mechanical dependence between its production characteristics and assigned local price, while the median limits sensitivity to outliers and reporting errors. Chicken and egg prices enter the empirical analysis separately to capture potentially different effects on household production and gender-specific labor allocation. As these measures reflect reported initial unit values rather than observed market transactions, they serve as proxies for local poultry prices at the beginning of the production period.

Figure 3 shows the distribution of local chicken and egg prices across villages and survey rounds. Chicken prices exhibit greater dispersion than egg prices, potentially reflecting differences in transportation costs, market access, and spatial market integration. These patterns are consistent with evidence of incomplete spatial price transmission in developing-country agricultural markets due to transportation costs and other market frictions (Rashid and Minot, 2010).

4.4 Outcome Measures and Control Variables

The empirical analysis combines individual labor outcomes with household-level poultry production and socioeconomic characteristics. Labor allocation is measured along both extensive and intensive margins. The extensive margin includes indicators for participation in poultry production and any income-generating activity during the preceding seven days, while the intensive margin captures total weekly working hours and hours devoted specifically to poultry production. These outcomes are examined separately for female and male household members aged 15–64 years, allowing changes in poultry-specific labor allocation to be distinguished from changes in overall labor supply.

Household poultry output is measured by the annual value of chicken and egg production. Production inputs include annual female and male family labor, expenditures on feed, medicine, and other variable inputs, the initial value of poultry stock, and productive household assets. These measures are used to estimate the household production function, from which gender-specific shadow wages and household shadow income are recovered for the labor-supply analysis.

All monetary variables are expressed to U.S. dollars at the average 2019 exchange rate (1 BDT = 0.01179 USD). Additional controls include household demographic composition, education, landholdings, productive assets, electricity and credit access, crop and livestock shocks, and measures of market access and local infrastructure. These characteristics account for observable differences in household resources, production conditions, and labor allocation opportunities. Detailed variable definitions are provided in Table A.2.

4.5 Descriptive Statistics

Table 1 presents descriptive statistics for the balanced panel of 1,464 poultry-producing households across three BIHS survey rounds, covering 257 villages. The sample includes 2,015–2,263 working-age women and 1,665–1,846 working-age men per round.

The statistics reveal pronounced gender differences in labor allocation. Women's poultry participation increased from 67.2% to 76.0% between the first and third rounds, while their average weekly poultry hours rose from 3.66 to 4.12. In contrast, men's poultry participation remained below 3%, with negligible working hours. Annual family labor inputs reinforce this specialization, averaging 230–282 hours for women compared with 5–13 hours for men. Despite women's dominant role in poultry production, men supplied substantially more labor overall, averaging 37–41 weekly hours compared with approximately 10.5 hours for women. While women's overall participation increased from 78.1% to 83.5%, their total working hours remained relatively stable, suggesting that increased poultry engagement may partly reflect labor reallocation across activities rather than an expansion in total labor supply. These contrasting patterns motivate the examination of gender-specific responses along both poultry-specific and overall labor margins.

Household characteristics also evolved over the survey period. Average years of education increased from 4.3 to 5.2 among women and from 4.7 to 5.5 among men. Although household size remained relatively stable, the number of children declined while the number of working-age adults increased, potentially affecting family labor availability. Average real household assets nearly doubled, from \$467 to \$911, while annual income excluding poultry increased from \$1,377 to \$2,138. Electricity access also expanded substantially, from 47.7% to 86.3%. These changes indicate improvements in household resources and infrastructure that may influence production decisions and labor allocation.

Poultry production expanded alongside these changes. The average annual real value of poultry output increased from \$26.4 to \$55.6, while the value of poultry inventory before annual production each survey round rose from \$15 to \$22.7. Over the survey period, average real chicken prices increased from \$1.77 to \$2.43 per bird, while egg prices rose from \$0.92 to \$1.39 per dozen. Figure 4 illustrates these changes across survey rounds, with error bars representing one standard deviation. Local prices also varied across villages within survey rounds (Figure 3). The figure shows both the upward trend in average poultry prices and the dispersion of local prices within each round, providing descriptive context for the subsequent analysis of household production and labor allocation. Together, the pronounced gender specialization in poultry production, differences in overall labor supply, and variation in local prices and household resources motivate the analysis of how poultry prices affect gender-specific shadow wages, household shadow income, and labor allocation.

5 Empirical Approach

The empirical strategy follows a nonseparable agricultural household framework to examine how local chicken and egg prices relate to gender-specific labor allocation. First, a household poultry production function is estimated to recover the marginal productivity of female and male family labor. Second, separability between production and household labor allocation decisions is tested. Third, the estimated production function is used

to construct gender-specific shadow wages and household shadow income. Finally, labor-supply equations examine the relationships between these measures and overall and poultry-specific labor participation and working hours.

5.1 Estimating the Household Production Function

The agricultural household model implies that under nonseparability, labor allocation depends on household-specific shadow wages rather than observed market wages. Because these shadow wages are not directly observed, they are recovered from an estimated household production function following Jacoby (1993), Skoufias (1994), and Makhdum and Kousar (2021). The estimated production technology also provides the information needed to recover household-specific shadow income, allowing the analysis to distinguish the substitution and income channels through which local poultry prices affect labor allocation.

Household poultry production is modeled using a Cobb–Douglas production function that uses the annual value of poultry output to family labor and other production inputs. The Cobb–Douglas specification is simple in estimation and interpretation (Abdulai and Regmi, 2000) and provides a parsimonious representation of household production technology and facilitates recovery of the marginal value products of family labor. Given the three-wave panel and moderate sample size, it also avoids the over-parameterization associated with more flexible functional forms. Therefore the estimation primarily proceed with the Cobb-Douglas form despite the well-known technological restrictions it imposes. The paper additionally estimates a translog production function, with the results reported in the Appendix.

For household h in survey round t, the poultry production function is estimated as

$$\ln V_{ht} = \alpha + \beta_f L^f_{hpt} + \beta_m L^m_{hpt} + \mathbf{Z}'_{ht}\boldsymbol{\delta} + \lambda_t + \mu_h + u_{ht}, \tag{11}$$

where V_{ht} denotes the annual real value of household poultry production, including both chicken and egg output; L^f_{hpt} and L^m_{hpt} denote annual female and male family labor devoted to poultry production (p), respectively; and $\mathbf{Z}_{ht}$ includes expenditures on feed, veterinary medicine, and other variable inputs, the initial value of poultry stock, productive household assets, and measures of market access and local infrastructure. Household fixed effects (μ) control for time-invariant differences in production technology, managerial ability, and other unobserved household characteristics, while survey-round fixed effects (λ) absorb common macroeconomic, technological, and policy shocks. Consequently, the production elasticities are identified from within-household variation in input use over time. Because output is measured as the annual value of production, the estimated marginal products correspond directly to the marginal value products of family labor, allowing household-specific shadow wages to be recovered without an additional output-price adjustment.

Several production inputs contain zero observations, particularly male family labor and expenditures on feed and veterinary services. To retain these observations while preserving a log-like functional form, all continuous production inputs are transformed using the Inverse Hyperbolic Sine (IHS) transformation, which is defined at zero and closely approximates the natural logarithm for larger values (Bellemare and Wichman, 2020). Details of the transformation are provided in Appendix 7.

Unlike the cross-sectional approaches of Jacoby (1993) and Makhdum and Kousar (2021), the three-wave BIHS panel allows household fixed effects to control for time-invariant unobserved heterogeneity, following Skoufias (1994). Such heterogeneity may include managerial ability, persistent poultry-rearing practices, and household-specific production conditions correlated with input choices. The fixed-effects estimator exploits within-household variation in production inputs over time, while survey-round effects account for common shocks. This approach mitigates omitted-variable bias arising from time-invariant heterogeneity but does not eliminate endogeneity associated with time-varying productivity shocks or simultaneous input decisions. The production function is estimated using the balanced panel of poultry-producing households, with standard errors clustered at the village level to account for correlated production shocks and local market conditions. The estimated production parameters are subsequently used to recover gender-specific shadow wages and household shadow income for the labor-supply analysis.

A further limitation concerns measurement error in self-reported family labor inputs, which may bias estimated marginal value products and the resulting shadow wages (Skoufias, 1994). Although household fixed effects eliminate time-invariant reporting differences, they do not address time-varying measurement error.

5.2 Estimating Shadow Wages and Shadow Income

Following Jacoby (1993), Skoufias (1994), and Le (2007), gender-specific shadow wages are derived from the marginal value products of family labor. Let $\widehat{V}_{ht}$ denote the predicted annual value of poultry production from Equation (11). Predicted rather than observed output is used to construct the shadow measures, reducing their sensitivity to transitory production shocks (Le, 2007; Makhdum and Kousar, 2021).

Accounting for the IHS transformation of labor inputs, the shadow wage for gender $g \in \{f, m\}$ is calculated as

$$\widehat{w}_{ht}^{g*} = \widehat{\beta}_g \frac{\widehat{V}_{ht}}{\sqrt{1 + (L_{hpt}^{g})^2}}, \tag{12}$$

where $\widehat{\beta}_g$ is the estimated coefficient on the labor input L_{hpt}^{g}. The resulting shadow wage measures the marginal value of an additional hour of family labor in poultry production. For sufficiently large labor inputs, Equation (12) approximates the conventional Cobb–Douglas marginal value product.

Household shadow income is constructed as

$$\widehat{y}_{ht}^{*} = \widehat{V}_{ht} - E_{ht} + B_{ht} + Y_{ht}^{NL}, \tag{13}$$

where E_{ht} represents variable poultry-production expenditures, B_{ht} new earnings from other production activities, and Y_{ht}^{NL} denotes income from non-poultry sources. This measure captures predicted net poultry production value combined with other household income, providing an indicator of household resources available for consumption.

The estimated shadow wages and income are subsequently incorporated into the labor-supply equations to examine their associations with gender-specific labor participation and working hours. Because these measures depend on estimated production parameters and household input choices, their use in the labor-supply

equations raises potential endogeneity concerns. The following subsections examine household separability and develop the instrumental-variable approach used to address these concerns.

5.3 Separability Tests

Under separability, competitive labor markets equate the marginal value product of family labor with the market wage, making household production decisions independent of preferences and labor endowments. Labor-market imperfections may instead create a deviation between shadow and market wages, leading to nonseparable production and labor-allocation decisions (Dillon et al., 2019). Because the degree and nature of nonseparability may vary across production activities, household characteristics, and labor-market conditions, two complementary tests are employed to assess this hypothesis.

5.3.1 Jacoby's Test of Separability

Following Jacoby (1993), the first test examines the relationship between recovered shadow wages and observed village-level agricultural wages. Separate equations are estimated for female and male labor:

$$\ln \widehat{w}^{*}_{g,ht} = \alpha_j + \beta_g \ln W^{M}_{g,vt} + \mathbf{Z}'_{ht}\boldsymbol{\gamma}_g + \lambda_t + \varepsilon_{g,ht}, \tag{14}$$

where $\widehat{w}^{*}_{g,ht}$ denotes the estimated shadow wage for gender $g \in \{f, m\}$, $W^{M}_{g,vt}$ is the corresponding village-level agricultural wage, $\mathbf{Z}_{ht}$ contains household controls, and λ_t represents survey-round fixed effects. Under separability, shadow wages equal market wages, implying $\alpha_j = 0$, $\beta_j = 1$, and $\boldsymbol{\gamma}_j = \mathbf{0}$. These restrictions are jointly evaluated using a Wald test.

The baseline specification is estimated using ordinary least squares (OLS), with survey-round fixed effects to account for common temporal changes. However, village-level agricultural wages may be endogenous if unobserved local labor demand, productivity shocks, or market conditions jointly influence market wages and household-specific shadow wages, even after controlling for survey-round effects. As a supplementary analysis, an instrumental variables (IV) specification is estimated using village-level non-agricultural wages and distance to the nearest market as excluded instruments for agricultural wages (Makhdum and Kousar, 2021). The IV estimates are interpreted cautiously, as these instruments may also be associated with household production conditions and returns to family labor. Because agricultural wages are observed only in the second and third BIHS rounds, the separability estimates drop first round observations.

5.3.2 Le's Generalized Test of Separability

As a complementary approach, the generalized test of Le (2010) examines whether household demographic characteristics influence production decisions after controlling for market wages and production inputs. The estimating equation is

$$\ln\left(\frac{V_{ht}}{L_{hpt}}\right) = \alpha + \sum_{g\in\{f,m\}} \beta_g \ln W^{M,g}_{vt} + \mathbf{Z}'_{ht}\boldsymbol{\gamma} + \mathbf{H}'_{ht}\boldsymbol{\delta} + \lambda_t + \varepsilon_{ht}, \tag{15}$$

where V_{ht} is the annual value of poultry output, L_{hpt} denotes total family poultry labor, $\mathbf{Z}_{ht}$ includes other production inputs, and $\mathbf{H}_{ht}$ contains household demographic characteristics excluded from production

decisions under separability. The primary implication of separability is that household demographics have no independent effect on production, yielding the null hypothesis $H_0 : \boldsymbol{\delta} = \mathbf{0}$. The restrictions are evaluated using a Wald test. Both OLS and IV specifications are estimated, using the same excluded instruments as in Jacoby's test. Jacoby's test evaluates the equality of shadow and market wages, whereas Le's test examines whether household demographics independently predict production outcomes conditional on market wages and inputs. Rejection of either test's null provides evidence against separability, motivating the use of household-specific shadow wages in the subsequent labor-supply analysis.

5.4 Estimating the Labor-Supply Function

The final stage examines how household-specific shadow wages and shadow income influence individual labor allocation. The empirical specification distinguishes the wage and income-related channels through which local poultry prices may affect labor allocation. For individual i in household h, village v, and survey round t, the labor-supply equation is specified as

$$L^g_{ihvt} = \alpha^g + \delta^g \ln \widehat{w}^{*g}_{ht} + \phi^g \ln \widehat{y}^*_{ht} + \mathbf{X}^{g\prime}_{ihvt}\boldsymbol{\theta}^g + \lambda^g_t + \varepsilon^g_{ihvt}, \qquad \text{for } g \in (f, m) \tag{16}$$

Here, g denotes gender, $\widehat{w}^g_{ht}$ represents the corresponding household-specific shadow wage, and $\widehat{y}ht$ denotes household shadow income. The dependent variable, L^gihvt, represents one of four labor outcomes: participation in poultry production, weekly poultry working hours, participation in any income-generating activity, or total weekly working hours. The vector $\mathbf{X}^g_{ihvt}$ includes individual and household characteristics, while λ^g_t denotes survey-round fixed effects. Separate equations are estimated for women and men aged 15–64 years.

Each equation includes the shadow wage corresponding to the individual's gender, reflecting the marginal return to their own labor in household poultry production. This specification focuses on own-shadow-wage responses, motivated by the gendered division of poultry production responsibilities and differences in employment opportunities outside the household. Household shadow income enters both equations to account for the income-related channel of labor allocation. Although female and male labor decisions may be interdependent, the specification does not separately identify cross-gender shadow-wage effects. Conditional on the instrumental-variable identification assumptions, δ^g captures the effect of the corresponding shadow wage on labor supply, holding household shadow income constant, while ϕ^g captures the effect of shadow income, holding the corresponding shadow wage constant.

Because shadow wages and shadow income are derived from household production decisions, they may be correlated with unobserved determinants of labor supply, potentially leading to endogenous regressors (Jacoby, 1993; Le, 2007; Skoufias, 1994). To address this concern, the labor-supply equations are estimated using limited-information maximum likelihood (LIML), which is generally less susceptible to finite-sample bias than conventional two-stage least squares (2SLS) in the presence of weak or numerous instruments (Andrews et al., 2019; Stock et al., 2002). The excluded instruments include local chicken and egg prices, pre-production household assets and initial poultry stock, and their interactions. The identification strategy exploits variation in local poultry prices and differences in households' productive endowments, which

may generate heterogeneous responses in the marginal returns to family labor and household income. Household assets and initial poultry stock are measured before the production cycle in each survey round, reducing concerns about their simultaneous determination with current poultry production. Nevertheless, identification requires that the instruments affect labor supply only through the endogenous shadow wages and shadow income, conditional on the included controls. Instrument relevance is evaluated using the Sanderson–Windmeijer conditional F-statistics and Kleibergen–Paap identification diagnostics, while the Hansen J test assesses the overidentifying restrictions.

The identification strategy exploits variation in local poultry prices and differences in households' productive endowments. Changes in chicken and egg prices affect the value of poultry output and the marginal returns to family labor, while initial poultry stock and productive assets influence the scale and capacity of household production. Consequently, households with different initial endowments may experience heterogeneous changes in shadow wages and production income in response to local price variation. The interaction terms capture this differential exposure, providing additional variation for identifying the two endogenous shadow measures.

The validity of the IV strategy rests on instrument relevance and the exclusion restriction. Relevance requires that local chicken and egg prices and their interactions with household productive endowments predict shadow wages and shadow income, conditional on the included covariates and survey-round effects. The exclusion restriction assumes that these instruments influence individual labor allocation through household-specific shadow wages and shadow income rather than through independent channels. Several features of the empirical setting support the plausibility of this assumption. First, local poultry prices are constructed from village-level measures that exclude each household's own reported valuation, reducing mechanical dependence between the instruments and household production decisions. Second, prices are measured at the beginning of the annual production period, preceding both annual poultry production and the subsequent measurement of individual labor outcomes, which limits contemporaneous reverse causality. Third, interactions between local prices and predetermined productive endowments capture differences in households' capacity to translate market incentives into poultry production returns. Conditional on the main effects of prices and endowments, these interactions provide variation in predicted shadow wages and shadow income arising from differential exposure to common local price conditions. Nevertheless, the exclusion restriction could be violated if local prices influence labor allocation through production adjustments not fully captured by the shadow measures, or if productive endowments are correlated with unobserved determinants of labor supply. The estimates are therefore interpreted conditional on the maintained exclusion restriction. Instrument relevance and identification are assessed using first-stage estimates, conditional first-stage diagnostics, and weak-identification tests. As a robustness check, an alternative specification controls for the constituent price and endowment variables and uses only their interactions as excluded instruments, assessing whether the estimated relationships persist after accounting for their direct effects. Standard errors are clustered at the village level to account for within-village correlation in production shocks and labor-market conditions. First-stage estimates and identification diagnostics are reported alongside the 2SLS-LIML results.

6 Results

This section presents the empirical results in five steps. It begins with the estimated production technology, then examines the separability assumption before presenting the instrumental-variable evidence on how local poultry prices affect household labor allocation.

6.1 Production Function Estimates

Table 2 presents the household poultry production-function estimates using pooled ordinary least squares (OLS), household fixed effects (FE), and random effects (RE). The discussion focuses on the FE estimates in Column (2). The Hausman test favors FE over RE ($\chi^2 = 13.463$, $p = 0.001$), consistent with correlation between household-specific unobserved heterogeneity and production inputs. The FE estimator accounts for persistent differences in managerial ability and production conditions, although potential endogeneity from time-varying productivity shocks and simultaneous input choices remains. Estimates from the alternative translog specification are reported in Table A.3.

The FE estimates indicate that both female and male family labor contribute positively and significantly to household poultry production. The coefficient on female labor (0.078, $p < 0.01$) is larger than that on male labor (0.028, $p < 0.05$), suggesting a stronger output response to comparable changes in transformed female labor inputs. This pattern is consistent with evidence from Bangladesh that women's labor contributes significantly to agricultural productivity (Rahman, 2010) and that women undertake most routine poultry-rearing activities, including feeding, cleaning, and bird management (Monir and Shuva, 2016; Monira et al., 2025). The gender difference may therefore reflect the task specialization characteristic of smallholder poultry production. However, the coefficients do not directly measure gender differences in marginal value products, which also depend on labor-input levels (Quisumbing, 1996).

Among other production inputs, initial poultry stock has the largest estimated coefficient (0.263, $p < 0.01$), highlighting the importance of flock size in determining household poultry output. Households with larger initial poultry stocks generally have greater productive capacity, allowing them to generate higher output values. Expenditures on feed, veterinary medicine, and other variable inputs are also positively associated with production (0.104, $p < 0.01$), consistent with the role of purchased inputs in supporting poultry growth, health, and productivity. Productive household assets have a smaller but statistically significant positive coefficient (0.034, $p < 0.01$), suggesting that access to complementary productive resources is associated with higher poultry output.

Household-head literacy is positively associated with poultry production (0.209, $p < 0.05$), consistent with the potential role of human capital in improving production management and the adoption of better husbandry practices (Kousar and Abdulai, 2016). In contrast, the gender and age of the household head are not statistically significant, providing no evidence of systematic differences in poultry output associated with these characteristics after controlling for other inputs and household fixed effects. Distance to the nearest market is negatively associated with poultry output ($\beta = -0.014$, $p < 0.10$), consistent with evidence that geographic isolation and transaction costs can constrain agricultural production by limiting market access and weakening production incentives (Key et al., 2000; Stifel and Minten, 2008). Electricity access and crop or

livestock shocks are not statistically significant.

Overall, the results highlight the importance of female family labor, initial poultry stock, and complementary production inputs in determining household poultry output. The gender-specific labor coefficients, together with the substantial differences in female and male labor inputs, underscore the need to recover household-specific marginal value products rather than relying solely on observed market wages. These estimates provide the basis for constructing gender-specific shadow wages and household shadow income for the subsequent analysis of labor allocation.

6.2 Household-Specific Shadow Wages and Shadow Income

Table 3 reports the estimated gender-specific shadow wages and household shadow income derived from the Cobb–Douglas and translog production functions, together with observed agricultural market wages. Following Jacoby (1993) and Skoufias (1994), shadow wages are recovered from the estimated marginal value products of family labor and represent the implicit returns to labor allocated to household production. The discussion focuses on the Cobb–Douglas estimates in Panel A, with the translog estimates in Panel B used to assess sensitivity to the functional-form assumption.

The Cobb–Douglas estimates reveal substantial gender differences in shadow wages. Average female shadow wages increase from \$0.025 per hour in the first round to \$0.031 in the second and \$0.036 in the third, while male shadow wages rise from \$0.508 to \$0.726 and \$0.968, respectively. Male shadow wages are considerably higher despite the larger estimated production elasticity of female labor. This differential partly reflects the gendered allocation of labor in household poultry production, where women undertake most routine management activities and contribute substantially more labor hours, while male involvement is often limited. Under the Cobb–Douglas specification, estimated marginal value products depend on both production elasticities and labor inputs; consequently, very low male labor hours can generate high model-implied shadow wages. Previous research documents gender specialization in agricultural production and differences in the marginal productivity of male and female labor, although the magnitude of these differences can be sensitive to the measurement of labor and output (Jacoby, 1991, 1993; Skoufias, 1994). These estimates should not be interpreted as evidence of intrinsically lower female productivity or as directly comparable measures of market earnings.

Estimated shadow wages increase across all three survey rounds for both genders, although the increase is more pronounced for male labor. These changes may reflect differences in poultry output values, input use, and the allocation of family labor over time. Household shadow income per capita also increases, from \$338 in the first round to \$408 in the second and \$522 in the third. The increase suggests an improvement in estimated household income over the study period, although the descriptive statistics do not establish the relative contributions of poultry production, non-poultry earnings, or other household-level changes.

The translog estimates in Panel B produce a similar gender pattern, with female shadow wages remaining substantially below male shadow wages. Average female shadow wages increase from \$0.013 to \$0.024 per hour, compared with an increase from \$0.794 to \$1.651 for male labor. However, the estimated wage levels differ appreciably across functional forms, particularly for male labor, indicating that the recovered marginal

products are sensitive to the specification of the production technology. The translog male shadow-wage estimates are also based on smaller and varying samples, which limits direct comparisons of the reported means due to omitting the the negative estimated shadow wages. In contrast, estimated shadow income per capita is nearly identical across the two specifications, ranging from approximately $338 in the first round to $522 in the third.

Panel C presents observed agricultural market wages for the second and third rounds. Female market wages increase from $0.271 to $0.363 per hour, while male market wages rise from $0.420 to $0.568. Estimated female shadow wages are substantially below observed female market wages, whereas male shadow wages exceed their corresponding market wages. This pattern may reflect gender specialization in household poultry production, where women undertake most routine activities while men's labor involvement is relatively limited. Women's greater allocation of time to poultry production, together with constraints on accessing alternative employment, may contribute to lower marginal returns relative to observed market wages. Conversely, very low male labor inputs can generate relatively high estimated marginal returns under the Cobb–Douglas production function. In a separable agricultural household model with competitive and accessible labor markets, the marginal value product of family labor is expected to equal the relevant market wage. Deviations from this equality may arise from labor-market constraints, transaction costs, and other imperfections that link production and labor-supply decisions (Benjamin, 1992; Jacoby, 1993; Skoufias, 1994). However, the magnitude of the estimated differences may also reflect functional-form assumptions and measurement error in labor inputs. Moreover, the reported averages are based on different samples and do not constitute a formal test of wage equality or production–consumption separability.

Overall, the results highlight substantial gender differences in the implicit returns to family labor in poultry production. They also demonstrate that the estimated opportunity cost of household labor can differ considerably from observed agricultural wages. These patterns motivate the formal separability tests and subsequent analysis of gender-specific labor allocation using estimated shadow wages and household income.

6.3 Separability Tests

The separable agricultural household model assumes that households can allocate family labor between agricultural production and market employment at prevailing market wages, allowing production decisions to be made independently of household labor-supply preferences and demographic characteristics. When labor markets are imperfect, however, observed wages may differ from the opportunity cost of family labor, making production and labor-allocation decisions interdependent (Benjamin, 1992; Jacoby, 1993; Skoufias, 1994). To assess these implications, the analysis employs the complementary separability tests proposed by Jacoby (1993) and Le (2010), as described in Section 5.3. Ordinary least squares (OLS) serves as the primary specification. As a supplementary analysis, instrumental-variable (IV) estimates are reported to examine sensitivity to potential endogeneity of observed village-level agricultural wages. Such endogeneity may arise if unobserved local labor demand, agricultural productivity, or market conditions jointly influence agricultural wages and household-specific shadow wages. Although survey-round fixed effects absorb common temporal shocks, they do not necessarily account for these local factors. The IV results are therefore interpreted cautiously, conditional on the validity of the excluded instruments.

Table 4 presents the results of Jacoby's test, which examines whether estimated household-specific shadow wages are consistent with observed agricultural market wages. The joint Wald tests based on the OLS specifications reject the restrictions implied by separability for both female and male labor ($p < 0.001$). The supplementary IV specifications yield the same qualitative conclusion, although their interpretation depends on the validity of the excluded instruments. First stage results of the IV is shows in Table A.4. These findings indicate that estimated shadow wages do not satisfy the restrictions implied by equality between household-specific returns to family labor and observed agricultural wages. Such discrepancies are consistent with labor-market imperfections, including transaction costs and limited access to wage employment, that may prevent households from equating marginal returns to family labor with prevailing market wages (Jacoby, 1993; Skoufias, 1994).

Table 5 reports the results of Le's generalized separability test (first stage result is shown in Table A.5, which examines whether household demographic characteristics remain associated with production decisions after accounting for observed market wages and other production-related factors. Under separability, household composition should not independently influence production decisions conditional on relevant market conditions and productive resources. The OLS-based joint Wald tests reject the imposed separability restrictions ($p < 0.001$), and the supplementary IV specifications yield the same conclusion. These findings suggest that household characteristics remain relevant to production decisions, consistent with the nonseparable agricultural household framework. The agreement between OLS and IV results indicates that the qualitative conclusion is unchanged across the reported specifications, although it does not establish the validity of the instruments.

Taken together, the two tests reject key restrictions of the separable agricultural household model among poultry-producing households in rural Bangladesh. Jacoby's test indicates that estimated shadow wages do not satisfy the restrictions implied by observed agricultural market wages, while Le's test shows that household demographic characteristics remain associated with production decisions conditional on market wages and other controls. These findings are consistent with previous evidence of nonseparability in developing-country agriculture (Makhdum and Kousar, 2021; Skoufias, 1994) and motivate the use of household-specific shadow wages in the subsequent analysis of gender-specific labor allocation. Although the results are consistent with interdependence between production and labor-allocation decisions, the tests do not identify the particular labor-market imperfections responsible for these relationships.

6.4 Household Labor Supply Estimates

Table 7 presents the Instrumental Variable (IV) estimates of gender-specific labor-supply responses to household shadow wages and shadow income, while Table 6 reports the corresponding first-stage estimates and Table A.9 shows the reduced form of the IV estimation. Together, these results help explain how changes in local chicken and egg prices may influence household labor allocation through household-specific production returns and income. In a nonseparable agricultural household framework, changes in output prices can affect the marginal value product of family labor and household income, potentially generating different labor-supply responses through shadow wages and shadow income (Jacoby, 1993; Singh et al., 1986; Skoufias, 1994). To examine these mechanisms, the analysis considers four outcomes separately for

women and men: participation in poultry production, weekly poultry working hours, overall participation in income-generating activities, and total weekly working hours. Distinguishing poultry-specific from overall labor supply is particularly important because higher returns to poultry production may change the allocation of labor across activities without necessarily increasing total working time.

First-stage results and identification. The first-stage estimates in Table 6 indicate that the relationships between local poultry prices, household shadow wages, and shadow income vary with productive endowments. For male shadow wages, the coefficient on chicken prices is positive (1.262, $p < 0.05$), while its interaction with household assets is negative (−0.255, $p < 0.05$), suggesting that the association between chicken prices and male shadow wages weakens as household assets increase. One possible explanation is that better-endowed households have greater flexibility in organizing poultry production and accessing alternative inputs or hired labor, making the marginal return to additional male family labor less sensitive to chicken price changes. In contrast, households with fewer productive resources may rely more heavily on male family labor for activities such as input procurement, transportation, and marketing, resulting in a stronger response of male shadow wages to chicken prices. Similarly, the negative interaction between egg prices and initial poultry capital (−0.221, $p < 0.10$) may reflect differences in how households with varying initial production capacity respond to market incentives. Initial poultry capital is positively associated with both female and male shadow wages, with coefficients of 0.172 and 0.252, respectively ($p < 0.01$), while household assets are positively associated with male shadow wages (0.216, $p < 0.01$). These patterns suggest that productive endowments may shape how local poultry prices translate into household-specific marginal returns to labor. Nevertheless, the underlying mechanisms cannot be identified directly from the first-stage estimates, and the interaction coefficients should be interpreted as conditional associations rather than evidence of particular production adjustments.

The Sanderson–Windmeijer conditional F-statistics are 35.374 and 21.588 for female shadow wages and income, respectively, compared with 13.859 and 14.014 for the corresponding male variables. These statistics assess instrument relevance for each endogenous variable conditional on the other endogenous regressor (Sanderson and Windmeijer, 2016). The Kleibergen–Paap LM tests reject underidentification in all specifications ($p < 0.001$). However, the Kleibergen–Paap rk Wald F-statistic is lower for men (8.659) than for women (20.378), indicating greater concern about weak identification in the male specifications. Although LIML can reduce finite-sample bias relative to conventional 2SLS under weak identification, it does not eliminate this concern (Andrews et al., 2019; Stock et al., 2002). The Hansen J test does not reject the overidentifying restrictions in seven of the eight labor-supply equations but rejects them for female overall working hours ($p = 0.008$). These diagnostics do not establish the exclusion restriction, which requires that local poultry prices and household productive endowments affect labor supply only through the instrumented shadow wages and shadow income, conditional on the included controls. Accordingly, the results are interpreted subject to this identifying assumption.

For comparison, Appendix Table A.6 reports ordinary least squares (OLS) estimates of the labor-supply equations. The OLS and IV-LIML estimates differ in magnitude and statistical significance, indicating that the estimated relationships are sensitive to the treatment of household-specific shadow wages and shadow income

as endogenous variables. Because these measures are recovered from household production decisions, they may be correlated with unobserved determinants of individual labor allocation, including labor preferences, productive ability, and household-specific constraints. In addition, measurement error in the estimated shadow measures may affect the OLS coefficients. These considerations motivate the IV-LIML approach, although differences between OLS and IV-LIML do not independently establish endogeneity or demonstrate the validity of the excluded instruments.

Female labor-supply responses. Panel A of Table 7 shows that female shadow wages are positively and significantly associated with poultry participation and weekly poultry working hours. The estimated coefficient on poultry participation is 0.075 ($p < 0.01$), implying that a 10% increase in the female shadow wage is associated with approximately a 0.75 percentage-point increase in the probability of participating in poultry production. Relative to the sample participation rate of 72%, this represents an increase of approximately 1%. The corresponding coefficient on weekly poultry hours is 0.947 ($p < 0.01$), implying an increase of approximately 0.09 hours, or 5.4 minutes, per week for a 10% increase in the shadow wage. Relative to the sample mean of 3.98 weekly poultry hours, this represents an increase of approximately 2.3%. Although modest in absolute terms, the estimated hours response is larger relative to the low baseline level of poultry labor, suggesting that changes in household-specific returns are associated with adjustments in both women's participation and time devoted to poultry production.

These findings provide evidence that smallholder households respond to economic incentives in poultry production, although the magnitude of the response is limited. An increase in local chicken or egg prices that raises the marginal return to female poultry labor may encourage women to enter poultry production or devote additional time to existing activities. However, the estimated coefficients on female shadow wages are not statistically significant for overall labor-force participation (0.016) or total weekly working hours (-0.202). Thus, while higher shadow wages are associated with greater poultry-specific labor allocation, there is no statistically discernible increase in women's overall labor supply. The coefficient on total working hours is imprecisely estimated, however, and does not establish that overall labor supply remains unchanged. The contrast between poultry-specific and overall responses is consistent with the possibility that women reallocate time toward poultry production from other activities rather than substantially expanding their total working hours, although the estimates do not directly identify such adjustments.

The concentration of women's labor responses in poultry production is consistent with the gendered organization of smallholder poultry farming in Bangladesh, where women typically undertake routine feeding, cleaning, and bird care. Because these activities are integrated into household responsibilities, women may be able to make incremental adjustments to poultry labor in response to changing economic returns without substantially altering their overall employment patterns. However, expanding poultry production beyond small household flocks may require complementary investments in housing, purchased feed, veterinary inputs, and production-management skills. Competing household responsibilities and limited access to alternative labor arrangements may further constrain women's ability to expand their productive time. Although these mechanisms cannot be distinguished empirically, they offer plausible explanations for why poultry-specific labor responds to shadow wages without a clearly detectable expansion in overall labor supply. This pattern is

consistent with the nonseparable agricultural household framework, in which labor allocation responds to household-specific marginal returns under imperfect labor markets (Jacoby, 1993; Le, 2007; Skoufias, 1994).

The results, together, suggest that smallholder households treat poultry production as an economic activity responsive to changes in implicit labor returns, but the observed adjustments are relatively modest. This distinction is important for assessing whether expanding poultry markets can provide a pathway to substantial income growth for smallholder producers. If households primarily adjust labor within existing production activities, the benefits of favorable market conditions may remain limited without complementary investments that expand productive capacity or reduce constraints on women's time. While the present analysis does not identify the binding constraints, it highlights the potential importance of productive resources and gender-specific labor constraints in determining how smallholders benefit from poultry market opportunities.

Male labor-supply responses. Panel B presents a different pattern for men. The coefficients on male shadow wages are not statistically significant for poultry participation (0.004), weekly poultry working hours (0.068), overall participation (−0.008), or total weekly working hours (−1.713). These estimates provide no statistically detectable evidence that changes in male shadow wages are associated with adjustments in either poultry-specific or overall labor supply. This contrasts with the positive female poultry-labor responses and is consistent with the limited involvement of men in routine household poultry production. Only 1.4% of men participate in poultry activities, compared with 72% of women, while average weekly poultry hours are 0.06 for men and 3.98 for women. Consequently, even when changes in local poultry prices alter the implicit returns to male labor, the corresponding adjustments may be limited because poultry production accounts for a relatively small share of men's activities. Nevertheless, the estimates do not establish that male and female shadow-wage responses differ statistically, and the weaker identification diagnostics for men warrant caution.

Shadow-income responses. Changes in local poultry prices may also influence household labor allocation through shadow income. For poultry-producing households, an increase in chicken or egg prices can raise the value of poultry output and potentially increase household resources, although the net effect depends on production quantities, costs, and household characteristics. In standard agricultural household models, greater household resources may increase the demand for leisure or time devoted to nonmarket activities, reducing participation in income-generating work (Jacoby, 1993; Singh et al., 1986; Skoufias, 1994). Consistent with this possibility, the estimated coefficient on shadow income is negative and statistically significant for female poultry participation (−0.061, $p < 0.05$) and overall labor-force participation (−0.067, $p < 0.01$). A 10% increase in shadow income is associated with reductions of approximately 0.58 and 0.64 percentage points in these participation outcomes, respectively. The coefficient on weekly poultry hours is negative but statistically insignificant (−0.297), while the coefficient on total weekly hours is negative and significant at the 10% level (−1.732). The latter estimate requires particular caution because the Hansen J test rejects the overidentifying restrictions for female overall working hours.

For men, shadow income is negatively associated with overall labor-force participation (−0.078, $p < 0.05$), implying a reduction of approximately 0.74 percentage points for a 10% increase in shadow income. In contrast, the coefficient on total weekly working hours is positive (4.875, $p < 0.10$), corresponding to approximately 0.46 additional hours per week. The coefficients on male poultry participation (−0.015) and

poultry working hours (0.034) are not statistically significant. These results indicate that higher shadow income is associated with lower overall participation for both genders, but the working-hours responses differ. In particular, the positive male hours coefficient does not support a uniform reduction in labor supply as household resources increase. Because the working-hours outcomes include individuals with zero hours, these estimates cannot separately identify adjustments among existing workers from changes in labor-force participation.

The shadow-income findings suggest a potentially important countervailing mechanism through which local poultry prices may influence household labor allocation. For women, the estimated shadow-wage coefficient on poultry participation is positive (0.075), whereas the shadow-income coefficient is negative (-0.061), with both statistically significant. A linear combination test indicates that the sum of these coefficients is positive but small (0.014) and not statistically different from zero. This suggests that the opposing shadow-wage and shadow-income relationships may offset each other when both shadow measures increase proportionally. However, the coefficient sum does not represent the net effect of a poultry-price change, which depends on the relative responsiveness of shadow wages and shadow income to prices and may vary with household productive endowments. These findings are consistent with competing substitution- and income-related considerations in agricultural household models, although the estimated coefficients do not separately identify pure substitution and income effects.

Economic magnitude and interpretation. The estimated coefficients suggest economically meaningful but relatively modest labor-supply responses to changes in household-specific shadow wages and shadow income. A 10% increase in female shadow wages is associated with a 0.71 percentage-point increase in poultry participation, equivalent to approximately 1% of the baseline participation rate of 72%, and an additional 0.09 weekly poultry working hours, representing 2.3% of the sample mean of 3.98 hours. In contrast, a 10% increase in household shadow income is associated with a 0.58 percentage-point reduction in female poultry participation and reductions of 0.64 and 0.74 percentage points in overall labor-force participation among women and men, respectively. These findings are broadly consistent with earlier agricultural household studies documenting labor-supply responses to household-specific shadow wages and income (Jacoby, 1993; Skoufias, 1994). For example, Jacoby (1993) finds relatively small positive agricultural labor-supply elasticities with respect to shadow wages in rural Peru, ranging from approximately 0.02 to 0.08 across specifications. Although these estimates are not directly comparable because the present study examines poultry-specific participation and working hours rather than aggregate agricultural labor-supply elasticities, both studies indicate that household labor allocation responds to changes in implicit returns. The relatively modest absolute increase in poultry working hours in the present study is particularly relevant given the small scale of household poultry production, where adjustments may occur through incremental changes in routine management rather than substantial expansion of labor inputs.

The findings also highlight the potentially offsetting relationships between shadow wages, shadow income, and household labor allocation. Higher female shadow wages are associated with greater poultry participation and working hours, while higher household shadow income is associated with lower overall labor-force participation among both women and men. However, these coefficients describe responses to changes in the

shadow measures rather than the total effect of local poultry-price changes. The latter depends on how poultry prices influence shadow wages and income, with these relationships varying according to household assets and initial poultry capital. The concentration of the female shadow-wage response in poultry production, together with the absence of statistically discernible male shadow-wage responses, is consistent with gender specialization in smallholder poultry farming. Moreover, the lack of a statistically significant increase in women's overall working hours, although imprecisely estimated, suggests that greater poultry engagement may partly involve reallocating existing labor rather than expanding total labor effort. Such adjustments may reflect competing demands on women's time or limitations on expanding flock size without complementary capital, improved housing, feed, veterinary inputs, and technical skills. Although these constraints are not directly identified, the findings suggest that smallholder households respond to poultry market opportunities, but their capacity to expand production may depend on both existing productive assets and gender-specific labor constraints. By distinguishing poultry-specific from overall labor responses, the analysis extends the shadow-wage framework of Jacoby (1993) and Skoufias (1994) to a setting in which poultry production is highly gendered and opportunities for smallholder commercialization may be constrained.

6.5 Robustness Checks

This section examines the sensitivity of the baseline IV-LIML results to alternative sample definitions and production-function specifications. Specifically, the analysis considers a broader sample of poultry-producing households observed in at least two survey rounds and an alternative translog production function for recovering household-specific shadow wages and shadow income. These exercises assess the extent to which the estimated labor-supply relationships depend on the balanced-panel restriction and the assumed production technology.

First, Appendix Table A.8 reports IV-LIML estimates using the broader sample of poultry-producing households observed in at least two survey rounds. The positive relationship between female shadow wages and poultry-specific labor allocation remains broadly consistent with the baseline findings. However, unlike the baseline specification, the estimated female shadow-wage coefficient on overall labor-force participation is positive and statistically significant. This difference suggests that the relationship between female shadow wages and participation in income-generating activities beyond poultry may be sensitive to sample composition. Households observed in only two rounds may differ from those in the balanced panel in their productive resources, labor-market opportunities, or participation in poultry production. Although the broader-sample estimates provide some support for the positive poultry-specific response, the difference in overall participation indicates that the evidence on broader female labor-market engagement is less stable across sample definitions. Moreover, the comparison does not eliminate concerns about selection into poultry production or survey attrition.

Second, Appendix Table A.7 examines the sensitivity of the results to the functional form of the poultry production technology. Household-specific shadow wages and shadow income are recovered from a translog production function, which allows greater flexibility in input substitution and marginal productivity than the baseline Cobb–Douglas specification. The resulting IV-LIML estimates differ from the baseline, particularly for the relationship between female shadow wages and weekly poultry working hours. This difference indicates

that the estimated intensive-margin response is sensitive to the production-function specification used to construct shadow wages. In addition, the translog-based models exhibit weaker first-stage identification and less precise estimates, complicating the interpretation of differences across specifications. Consequently, the translog results do not provide strong confirmation of the baseline intensive-margin findings and instead highlight the importance of production-function assumptions for the estimated shadow-wage relationships.

Overall, the robustness exercises provide mixed evidence regarding the stability of the baseline findings. The broader sample generally supports the positive relationship between female shadow wages and poultry-specific labor allocation, although the statistically significant response in overall participation differs from the baseline estimates. However, including households observed in only two survey rounds may introduce additional concerns about sample selection, as these households may differ systematically in their poultry production decisions and labor allocation. The translog specification also produces materially different estimates for female poultry working hours, but its weaker first-stage identification raises concerns about the reliability of the corresponding IV-LIML estimates. These limitations provide reasons to prefer the baseline Cobb–Douglas IV-LIML specification, which uses a consistent three-round household sample and exhibits stronger first-stage identification. Nevertheless, the balanced-panel restriction does not eliminate selection concerns, and the baseline estimates remain conditional on the Cobb–Douglas production technology and the validity of the excluded instruments. Thus, while the alternative specifications reveal some sensitivity in the estimated labor-supply relationships, they do not provide equally well-identified alternatives to the baseline results.

7 Conclusion

This paper examines how local poultry prices relate to household-specific shadow wages, shadow income, and gendered labor allocation in rural Bangladesh. Using three rounds of the Bangladesh Integrated Household Survey and a nonseparable agricultural household framework, the analysis investigates how local chicken and egg prices are associated with implicit returns to family labor and how these returns relate to poultry-specific and overall labor supply among women and men.

The analysis yields three main findings. First, separability tests reject the restrictions implied by separable household production and labor-supply decisions, consistent with the importance of household-specific shadow wages under imperfect labor markets. Second, the relationships between local poultry prices and shadow wages and income vary with household assets and initial poultry capital. Thus, the same price change may generate different changes in implicit labor returns across households, reflecting differences in their productive resources and capacity to adjust production. Third, the IV-LIML estimates indicate that higher female shadow wages are associated with greater poultry participation and working hours, while the corresponding relationships with overall labor supply are not statistically significant in the baseline specification. A 10% increase in female shadow wages is associated with a 0.71 percentage-point increase in poultry participation and approximately 0.09 additional weekly poultry working hours. Male shadow wages show no statistically detectable relationships with labor-supply outcomes. In contrast, higher household shadow income is associated with lower overall labor-force participation among both women and men, suggesting potentially competing labor-supply incentives.

These findings have broader implications for women's capacity to respond to economic opportunities. Poultry production is typically undertaken near the household and can be combined with domestic responsibilities, making it relatively accessible to women facing mobility restrictions and competing demands on their time. The positive response of women's poultry labor to higher implicit returns suggests that women may respond to economic opportunities compatible with these constraints, even when changes in overall labor supply are not statistically detectable. This raises the possibility that other home-based enterprises and locally accessible employment opportunities could similarly support women's economic participation. Technologies and institutions that reduce barriers to market access, such as digital platforms, mobile financial services, and producer organizations, may further expand these opportunities. However, the study does not directly examine these activities or identify the effects of mobility and social constraints.

At the same time, the modest increase in poultry working hours suggests potential limits to expanding production through additional family labor alone. Greater participation in poultry markets may require complementary resources, including working capital, improved poultry housing, feed, veterinary services, and technical knowledge. Improving market access and production incentives may therefore be more effective when accompanied by measures that address productive constraints and women's competing time demands. These implications remain suggestive because the analysis does not directly estimate the effects of such interventions or establish whether increased poultry labor translates into higher output, profits, or women's welfare.

This study faces several limitations. The findings apply to poultry-producing households and may not generalize to other activities. The IV-LIML estimates depend on the validity and strength of the excluded instruments, while the recovered shadow measures are sensitive to assumptions about production technology. In particular, the translog specification yields different estimates for female poultry working hours and weaker first-stage identification. The analysis also does not directly observe labor reallocation, mobility constraints, or the welfare consequences of changes in women's labor allocation. Future research could examine whether new technologies and institutional arrangements enable women to translate responsiveness to household-based economic opportunities into sustained improvements in productivity, earnings, and economic well-being.

Overall, the findings suggest that women's labor allocation responds to household-specific returns in poultry production, even when corresponding changes in aggregate labor supply are not statistically detectable. Understanding how women and other smallholders can benefit from expanding agricultural markets therefore requires attention not only to price incentives but also to the productive resources and institutional conditions that shape their capacity to respond.

References

Abdulai, Awudu and Punya Prasad Regmi, "Estimating labor supply of farm households under nonseparability: empirical evidence from Nepal," *Agricultural Economics*, **2000,** *22* **(3), 309–320.**

Ahmed, Akhter, "Bangladesh integrated household survey (BIHS) 2011-2012," 2013.

__ , "Bangladesh Integrated Household Survey (BIHS) 2015," 2016.

__ , "Bangladesh Integrated Household Survey (BIHS) 2018-2019," 2020.

Andrews, Isaiah, James H. Stock, and Liyang Sun, "Weak Instruments in Instrumental Variables Regression: Theory and Practice," *Annual Review of Economics*, **2019,** *11*, **727–753.**

Barrett, Christopher B., Shane M. Sherlund, and Akinwumi A. Adesina, "Shadow wages, allocative inefficiency, and labor supply in smallholder agriculture," *Agricultural Economics*, **2008,** *38* **(1), 21–34.**

Beck, Ulrik, Saurabh Singhal, and Finn Tarp, "Commodity Prices and Intra-Household Labor Allocation," *American Journal of Agricultural Economics*, **March 2019,** *101* **(2), 436–454.**

Bellemare, Marc F. and Casey J. Wichman, "Elasticities and the Inverse Hyperbolic Sine Transformation," *Oxford Bulletin of Economics and Statistics*, **2020,** *82* **(1), 50–61.**

Benjamin, Dwayne, "Household Composition, Labor Markets, and Labor Demand: Testing for Separation in Agricultural Household Models," *Econometrica*, **1992,** *60* **(2), 287–322.**

Chowdhury, S. D., "Family poultry production in Bangladesh: is it meaningful or an aimless journey?," *World's Poultry Science Journal*, **2013,** *69* **(3), 649–665.**

Danzer, Alexander M. and Robert Grundke, "Export price shocks and rural labor markets: The role of labor market distortions," *Journal of Development Economics*, **2020,** *145*, **102464.**

de Janvry, Alain, Marcel Fafchamps, and Elisabeth Sadoulet, "Peasant Household Behaviour with Missing Markets: Some Paradoxes Explained.," *The Economic Journal*, **November 1992,** *101* **(3), 1400–1417.**

Deaton, Angus, "Quality, Quantity, and Spatial Variation of Price," *American Economic Review*, **1988,** *78* **(3), 418–430.**

__ , "Price Elasticities from Survey Data: Extensions and Indonesian Results," *Journal of Econometrics*, **1990,** *44* **(3), 281–309.**

Dillon, Brian, Peter Brummund, and Germano Mwabu, "Asymmetric non-separation and rural labor markets," *Journal of Development Economics*, **2019,** *139*, **78–96.**

DLS, "Livestock Economy at a Glance," 2025.

Dorward, Andrew, "Agricultural labour productivity, food prices and sustainable development impacts and indicators," *Food Policy*, **2013,** *39*, **40–50.**

Doss, Cheryl R, "Women and agricultural productivity: Reframing the issues," *Development Policy Review*, **2017,** *36* **(1), 35–50.**

Doss, Cheryl, Ruth Meinzen-Dick, Agnes Quisumbing, and Sophie Theis, "Women in agriculture: Four myths," *Global Food Security*, **2018,** *16*, **69–74.**

Ecker, Olivier and Andrew R. Comstock, "Dietary change and food demand in urbanizing Bangladesh," IFPRI discussion papers 2100, International Food Policy Research Institute (IFPRI) 2021.

FAO, *Small-scale poultry production: technical guide* **Food and Agriculture Organization of the United Nations 2004.**

Heintz, James, Naila Kabeer, and Simeen Mahmud, "Cultural norms, economic incentives and women's labour market behaviour: empirical insights from Bangladesh," *Oxford Development Studies*, **2018,** *46* **(2), 266–289.**

Hennessey, Mathew, Mehroosh Tak, Ambarish Karamchedu, and Ivo Syndicus, "Rise of the Golden Chicken: Tracking Bangladeshi Sonali Production amid Global Poultry Transformation," *Agriculture and Human Values*, **2026,** *43* **(2), 94.**

Høg, Erling, Guillaume Fournié, Md. Ahasanul Hoque, Rashed Mahmud, Dirk U. Pfeiffer, and Tony Barnett, "Avian Influenza Risk Environment: Live Bird Commodity Chains in Chattogram, Bangladesh," *Frontiers in Veterinary Science*, **2021,** *8*, **694753.**

Islam, S., J. Begum, M. I. Hossain, and M. Khatun, "Marketing of Live Poultry in Northwest of Dhaka City: A Value Chain Analysis," *American Journal of Food and Nutrition*, **2017,** *5* **(1), 28–40.**

Islam, S. M. F. and M. A. Jabbar, "Smallholder poultry model for poverty alleviation in Bangladesh: A review of evidence on impact," *Livestock Research for Rural Development*, **2005,** *17* **(10).**

Jacoby, Hanan G., "Productivity of Men and Women and the Sexual Division of Labor in Peasant Agriculture of the Peruvian Sierra," *Journal of Development Economics*, **1991,** *37* **(1–2), 265–287.**

__ , "Shadow Wages and Peasant Family Labour Supply: An Econometric Application to the Peruvian Sierra," *The Review of Economic Studies*, **1993,** *60* **(4), 903–921.**

Key, Nigel, Elisabeth Sadoulet, and Alain de Janvry, "Transactions Costs and Agricultural Household Supply Response," *American Journal of Agricultural Economics*, **2000,** *82* **(2), 245–259.**

Khan, M. M. H., M. Y. Miah, M. M. Bhuiyan, S. Begum, M. M. Hussain, and R. Khanum, "The Status of Homestead Poultry Production in Sylhet Region," *International Journal of Poultry Science*, **2006,** *5* **(11), 1092–1095.**

Kousar, Rakhshanda and Awudu Abdulai, "Off-farm work, land tenancy contracts and investment in soil conservation measures in rural Pakistan," *Australian Journal of Agricultural and Resource Economics*, **April 2016,** *60* **(2), None.**

Larive International B.V. and LightCastle Partners Ltd., "Poultry Sector Study Bangladesh (Revise 2023)," Commissioned Report, Netherlands Enterprise Agency (RVO) 2023. Commissioned by the Netherlands Enterprise Agency (RVO); Originally prepared for the Embassy of the Kingdom of the Netherlands in Bangladesh.

Le, Kien T., "Shadow Wages and Shadow Income in Farmers' Labor Supply Functions," *American Journal of Agricultural Economics*, **2007,** *91* **(3), 685–696.**

__ , "Separation Hypothesis Tests in the Agricultural Household Model," *American Journal of Agricultural Economics*, **2010,** *92* **(5), 1420–1431.**

Lederman, Daniel and Guido Porto, "The Price Is Not Always Right: On the Impacts of Commodity Prices on Households (and Countries)," *The World Bank Research Observer*, **2016,** *31* **(1), 168–197.**

Makhdum, Muhammad Sohail Amjad and Rakhshanda Kousar, "Gender in agriculture: Determinants of female labor supply decisions among rural households in the context of market imperfections in Pakistan," *Agricultural Economics*, **2021,** *52* **(2), 235–248.**

Miller, Grant and B. Piedad Urdinola, "Cyclicality, Mortality, and the Value of Time: The Case of Coffee Price Fluctuations and Child Survival in Colombia," *Journal of Political Economy*, **2010,** *118* **(1), 113–155.**

Monir, Mohammad Moniruzzaman and A. B. Shuva, "Gender issues in family poultry production in rural Bangladesh," *Family Poultry Communications*, **2016,** *25* **(1 & 2), 25–42.**

Monira, Kamrun Nahar, Nathu Ram Sarker, Sukumar Gosh, AHM Mohaiminul Islam, and Md. Ruhul Amin, "Socioeconomic and Demographic Determinants of Indigenous Chicken Rearing in Coastal Communities of Bangladesh," *Asian Journal of Agricultural Extension, Economics & Sociology*, **2025,** *43* **(9), 81–91.**

Moyen, N., G. Ahmed, S. Gupta, T. Tenzin, R. Khan, T. Khan, N. Debnath, M. Yamage, D. U. Pfeiffer, and G. Fournié, "A large-scale study of a poultry trading network in Bangladesh: implications for control and surveillance of avian influenza viruses," *BMC Veterinary Research*, **2018,** *14*, **12.**

Nath, Toma Deb, Arindam Biswas, and Sheikh Faruk Ahmed, "Livestock Farming Adoption in Rural Bangladesh: Determinants and Impacts on Household Income and Women's Dietary Diversity," *Food and Energy Security*, **2025,** *14* **(2), e70074. e70074 FES3-2024-11-0318.R1.**

National Institute of Population Research and Training (NIPORT) and ICF, "Bangladesh Demographic and Health Survey 2017–18," Final Report, NIPORT and ICF, Dhaka, Bangladesh, and Rockville, Maryland, USA 2020.

Nielsen, Hanne, Nanna Roos, and Shakuntala H. Thilsted, "The impact of semi-scavenging poultry production on the consumption of animal source foods by women and girls in Bangladesh," *The Journal of Nutrition*, **Nov 2003,** *133* **(11 Suppl 2), 4027S–4030S.**

Palacios-Lopez, Amparo, Luc Christiaensen, and Talip Kilic, "How much of the labor in African agriculture is provided by women?," *Food Policy*, **2017,** *67*, **52–63.**

Quisumbing, Agnes R., "Male-female differences in agricultural productivity: Methodological issues and empirical evidence," *World Development*, **1996,** *24* **(10), 1579–1595.**

Rahman, Sanzidur, "Women's Labour Contribution to Productivity and Efficiency in Agriculture: Empirical Evidence From Bangladesh," *Journal of Agricultural Economics*, **2010,** *61* **(2), 318–342.**

Rashid, Shahidur and Nicholas Minot, "Are Staple Food Markets in Africa Efficient? Spatial Price Analyses and Beyond," *Michigan State University, Department of Agricultural, Food, and Resource Economics, Food Security Collaborative Working Papers*, **01 2010.**

Rimi, Nadia Ali, Rebeca Sultana, Kazi Ishtiak-Ahmed, Md Zahidur Rahman, Marufa Hasin, M. Saiful Islam, Eduardo Azziz-Baumgartner, Nazmun Nahar, Emily S. Gurley, and Stephen P. Luby, "Understanding the Failure of a Behavior Change Intervention to Reduce Risk Behaviors for Avian Influenza Transmission Among Backyard Poultry Raisers in Rural Bangladesh: A Focused Ethnography," *BMC Public Health*, **2016,** *16* **(1), 858.**

Rosenzweig, Mark R., "Neoclassical Theory and the Optimizing Peasant: An Econometric Analysis of Market Family Labor Supply in a Developing Country," *The Quarterly Journal of Economics*, **February 1980,** *94* **(1), 31–55.**

Sanderson, Eleanor and Frank Windmeijer, "A Weak Instrument F-Test in Linear IV Models with Multiple Endogenous Variables," *Journal of Econometrics*, **2016,** *190* **(2), 212–221.**

Sarkar, K. and M. Golam, "A Move from Subsistence to Semi-Commercial Family Poultry Farming with Local Chickens: Effective Strategies for Family Poultry in Bangladesh," *World's Poultry Science Journal*, **2009,** *65* **(2), 251–259.**

Sattar, Abdullah Al, Rashed Mahmud, Md. Abu Shoieb Mohsin, Nurun Nahar Chisty, Md. Helal Uddin, Nusrat Irin, Tony Barnett, Guillaume Fournié, Eve Houghton, and Md. Ahasanul Hoque, "COVID-19 Impact on Poultry Production and Distribution Networks in Bangladesh," *Frontiers in Sustainable Food Systems*, **2021,** *5*, **714649.**

Shanta, I. S., M. A. Hasnat, N. Zeidner, E. S. Gurley, E. Azziz-Baumgartner, M. A. Y. Sharker, K. Hossain, S. U. Khan, N. Haider, A. A. Bhuyan, M. A. Hossain, and S. P. Luby, "Raising Backyard Poultry in Rural Bangladesh: Financial and Nutritional Benefits, but Persistent Risky Practices," *Transboundary and Emerging Diseases*, **2017,** *64* **(5), 1454–1464.**

Singh, Inderjit, Lyn Squire, and John Strauss, *Agricultural Household Models: Extensions, Applications, and Policy*, **Baltimore, MD: Johns Hopkins University Press, 1986.**

Skoufias, Emmanuel, "Using Shadow Wages to Estimate Labor Supply of Agricultural Households," *American Journal of Agricultural Economics*, **1994,** *76* **(2), 215–227.**

Sonaiya, Emmanuel Babafunso and S. E. J. Swan, *Small-Scale Poultry Production: Technical Guide* **number 1. In 'FAO Animal Production and Health Manual.', Rome: Food and Agriculture Organization of the United Nations, 2004.**

Stifel, David and Bart Minten, "Isolation and Agricultural Productivity," *Agricultural Economics*, **2008,** *39* **(1), 1–15.**

Stock, James H., Jonathan H. Wright, and Motohiro Yogo, "A survey of weak instruments and weak identification in generalized method of moments," *Journal of Business & Economic Statistics*, **2002,** *20* **(4), 518–529.**

Swaminathan, Madhura and Yoshifumi Usami, "Women's Role in the Livestock Economy," *Review of Agrarian Studies*, **2016,** *6* **(2), 123–134.**

Vijayamba, R. and Madhura Swaminathan, "Women's Work in Livestock Raising: Evidence from Time Use Surveys in India," *The Indian Journal of Labour Economics*, **2024,** *67* **(3), 681–704.**

Figures and Tables

Figure 1: Conceptual Framework: Poultry Prices and Household Labor Allocation

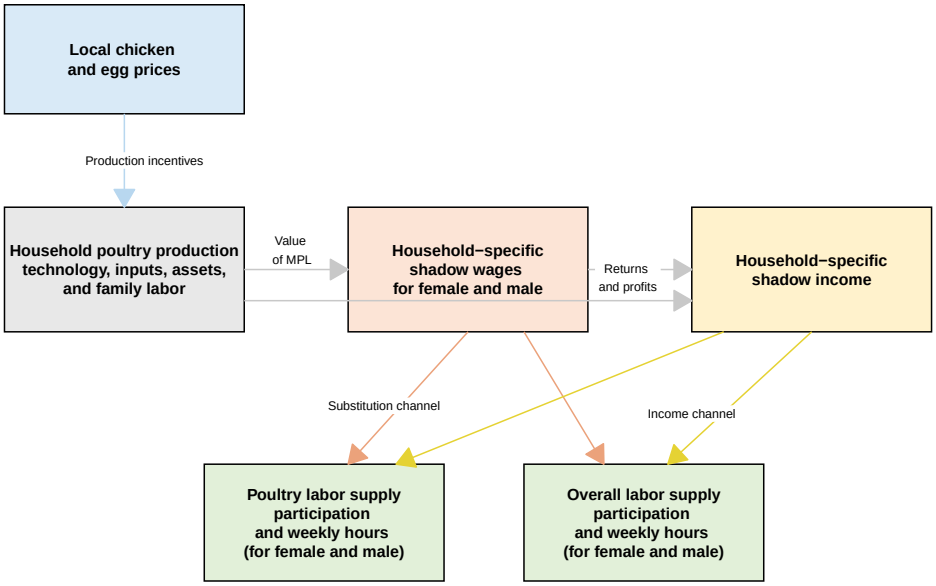
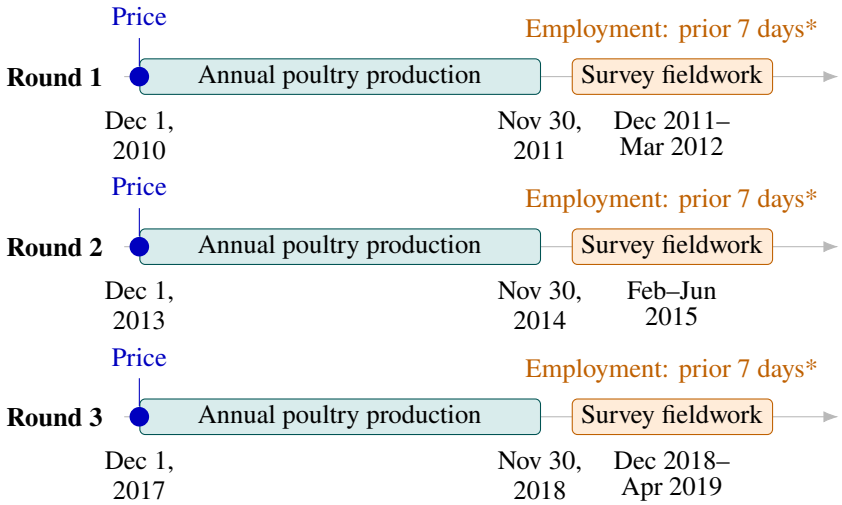
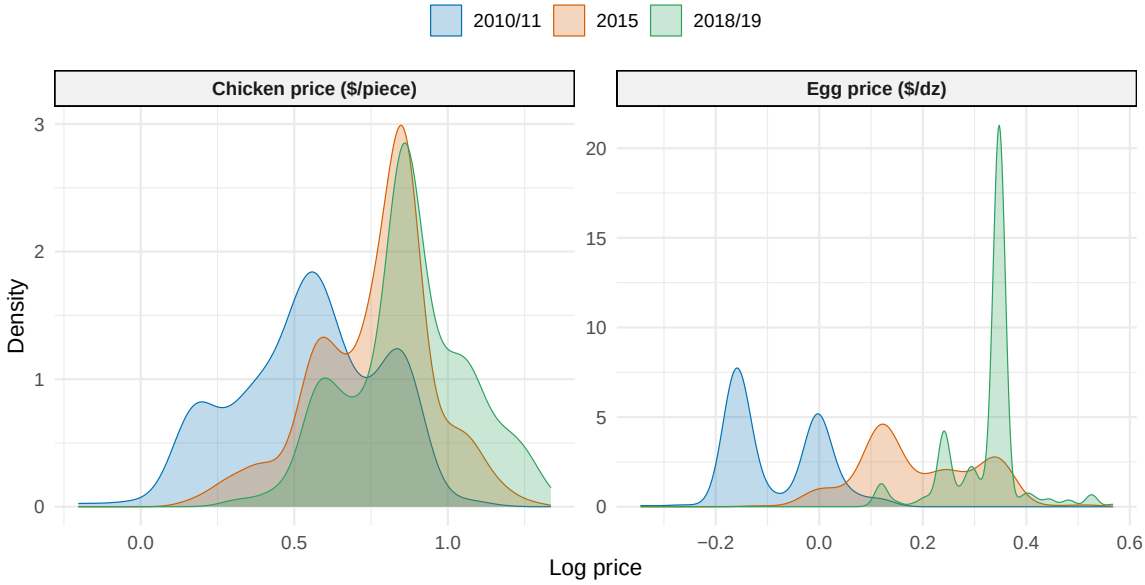


Figure 2: Timing of poultry prices, annual production, and employment outcomes across BIHS survey rounds



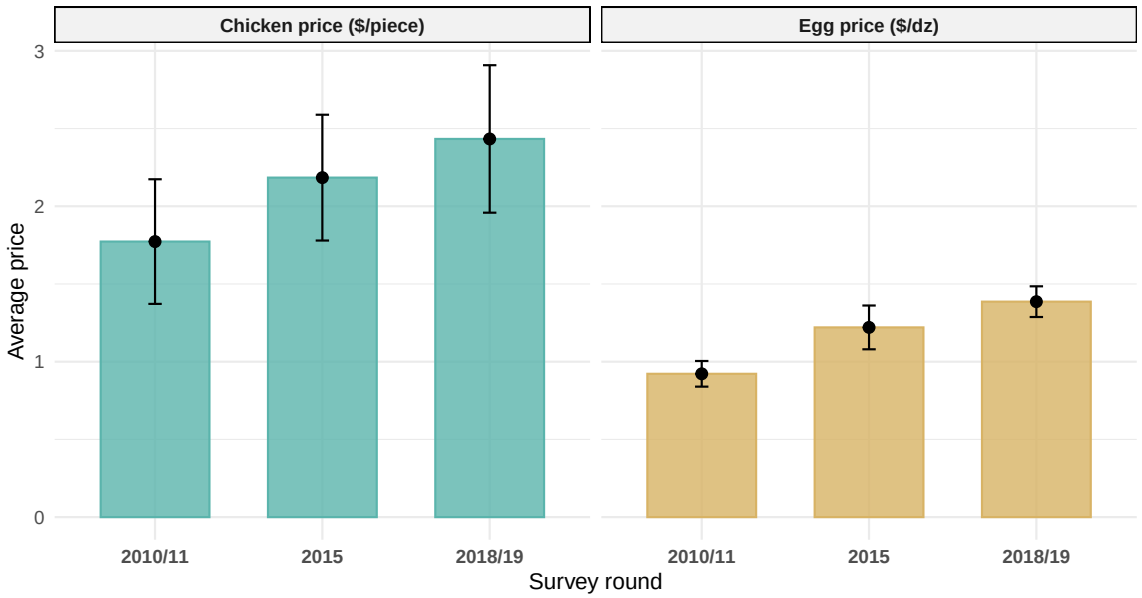
Notes: Blue dots indicate local chicken and egg prices at the beginning of each annual poultry production reference period. Green bars show the annual production reference periods. Orange bars indicate survey fieldwork windows, during which employment outcomes refer to the seven days preceding each household interview unless otherwise specified. The seven-day employment recall window varies by household and is not the full survey fieldwork interval. The horizontal spacing is schematic rather than proportional to elapsed time.

Figure 3: Distribution of chicken, egg prices over survey rounds



Notes: The figure presents kernel density estimates of log village-level chicken and egg prices across the three survey rounds for the balanced panel of poultry-producing households. Prices are expressed in US dollars per chicken and per dozen eggs, respectively, before logarithmic transformation. For each village and survey round, household-level log prices are averaged to obtain a single village-level observation, giving each village equal weight within a survey round. The density curves illustrate the distribution of local poultry prices across villages and changes in their dispersion over time. Prices are based on household-reported initial unit values at the beginning of each annual poultry production period.

Figure 4: Average chicken and egg prices



Notes: The figure presents average village-level chicken and egg prices across the three survey rounds for the balanced panel of poultry-producing households. Prices are expressed in US dollars per chicken and per dozen eggs, respectively. Village-level prices are first averaged within each survey round, and the reported means are calculated across villages, giving each village equal weight. Error bars represent one standard deviation above and below the mean across villages. Prices are based on household-reported initial unit values at the beginning of each annual poultry production period.

Table 1: Summary statistics for the estimation sample

	Round 1 (2010–11)		Round 2 (2015)		Round 3 (2018–19)	
Variable	Mean	SD	Mean	SD	Mean	SD
Panel A: Female characteristics						
Outcomes						
Poultry participation	0.672	0.470	0.722	0.448	0.760	0.427
Poultry weekly hours	3.660	4.067	4.105	4.287	4.124	4.974
Overall participation	0.781	0.414	0.802	0.398	0.835	0.371
Overall weekly hours	10.438	12.278	10.428	12.176	10.703	12.518
Characteristics						
Age	33.675	12.585	34.769	13.352	35.480	13.595
Married	0.782	0.413	0.761	0.426	0.768	0.422
Years of education	4.295	3.891	4.787	4.041	5.164	4.148
Panel B: Male characteristics						
Outcomes						
Poultry participation	0.008	0.088	0.007	0.084	0.028	0.164
Poultry weekly hours	0.023	0.316	0.077	1.237	0.082	0.742
Overall participation	0.883	0.322	0.846	0.361	0.861	0.346
Overall weekly hours	41.372	25.480	37.364	26.594	36.725	26.650
Characteristics						
Age	35.696	13.769	35.720	14.690	35.156	14.636
Married	0.732	0.443	0.683	0.465	0.655	0.475
Years of education	4.701	4.282	5.130	4.380	5.493	4.496
Panel C: Household characteristics						
Household size	4.361	1.513	4.482	1.655	4.539	1.826
Number of children	1.643	1.173	1.546	1.148	1.457	1.180
Number of adult males	1.137	0.724	1.240	0.823	1.261	0.881
Number of adult females	1.376	0.625	1.489	0.727	1.546	0.777
Number of elderly members	0.204	0.446	0.206	0.448	0.275	0.506
Average years of adult education	4.205	3.205	4.621	3.124	5.042	3.094
Household assets (usd/year)	466.771	876.365	660.901	949.855	910.955	1150.516
Household income w/o poultry (usd/year)	1377.103	2434.533	1708.763	2034.987	2138.335	1983.307
Homestead land (decimals)	11.521	13.709	11.568	13.215	12.017	15.252
Agricultural land (decimals)	99.437	144.790	105.107	178.547	101.575	139.409
Distance to main road (km)	1.976	3.452	1.784	3.722	2.206	3.202
Distance to local market (km)	0.653	0.714	0.519	0.640	0.635	1.474
Distance to district market (km)	1.821	1.658	1.727	1.641	1.916	2.228
Access to electricity	0.477	0.500	0.578	0.494	0.863	0.344
Experienced crop/livestock shock	0.130	0.337	0.075	0.264	0.125	0.331
Household has loan	0.699	0.459	0.742	0.438	0.754	0.431
Household produce crop	0.848	0.359	0.609	0.488	0.615	0.487
Panel D: Poultry production						
Value of poultry production (usd/year)	26.399	33.642	34.887	35.904	55.581	48.801
Initial poultry asset (usd)	14.995	16.404	16.681	21.054	22.738	39.637
Cost for feeding, medicine and other (usd)	2.878	5.454	3.872	7.558	2.497	13.254
Family female labor (hours/year)	229.689	152.775	281.867	151.566	236.872	132.775
Family male labor (hours/year)	5.167	31.570	9.736	52.985	13.250	41.030
Panel E: Local poultry prices						
Chicken price (usd/piece)	1.773	0.401	2.184	0.405	2.433	0.474
Egg price (usd/dz)	0.922	0.083	1.221	0.141	1.386	0.099
Number of female	2015		2180		2263	
Number of male	1665		1816		1846	
Number of households	1464		1464		1464	
Number of local markets (village)	257		257		257	

Notes: The table reports descriptive statistics for the balanced-panel estimation sample consisting of households engaged in poultry production in all three survey rounds. Individual characteristics are calculated for working-age household members aged 15–64, while household characteristics are reported at the household level. Monetary values are expressed in constant 2019 U.S. dollars. Village-level poultry prices represent the average local market prices faced by households in each survey round.

Table 2: Cobb-Douglas production function estimates for household poultry production

	Log of poultry output value		
	OLS	**Fixed Effect**	Random Effect
	(1)	(2)	(3)
Log of family female labor (hours/year)	0.094***	0.078***	0.093***
	[0.019]	[0.021]	[0.019]
Log of family male labor (hours/year)	0.028**	0.028**	0.028**
	[0.012]	[0.013]	[0.012]
Log of cost for feeding, medicine and others	0.070***	0.104***	0.072***
	[0.015]	[0.017]	[0.014]
Log of initial flock size	0.340***	0.263***	0.336***
	[0.015]	[0.018]	[0.015]
Log of productive asset	0.031***	0.034***	0.031***
	[0.008]	[0.013]	[0.008]
Household head is male (=1)	0.020	-0.080	0.019
	[0.039]	[0.074]	[0.039]
Age of the household head (years)	0.002	0.002	0.002
	[0.001]	[0.003]	[0.001]
Household head can read and write (=1)	0.076**	0.209**	0.077**
	[0.031]	[0.092]	[0.031]
Crop and Livestock shock	-0.017	0.009	-0.016
	[0.044]	[0.057]	[0.044]
Distance to nearest market (km)	0.005	-0.014*	0.004
	[0.008]	[0.008]	[0.008]
Electricity access (=1)	0.016	-0.009	0.015
	[0.038]	[0.055]	[0.038]
Constant	0.934***	1.234***	0.949***
	[0.125]	[0.195]	[0.125]
Observations	4392	4392	4392
Within R^2	0.298	0.288	0.301
F(Fixed effect)/Wald(Random effect)		67.560	1635.451
F(Fixed effect)/Wald(Random effect): *p-value*		0.000	0.000
Hausman test χ^2		13.463	
Hausman test p-value		0.001	

Notes: The production function is estimated using Cobb-Douglas production function. The dependent variable is the log of the annual value of poultry output (2019 USD). Family male labor, family female labor, livestock capital, production expenditures, and productive assets are measured in logarithms. All specifications include survey-round fixed effects. Columns (2) and (3) report household fixed-effects (FE) and random-effects (RE) estimates, with standard errors clustered at the village level reported in brackets. The Hausman test evaluates the null hypothesis that the random-effects estimator is consistent; rejection of the null supports the Kuse of the fixed-effects estimator. The Fixed Effect production-function estimates are subsequently used to recover household-specific shadow wages and shadow income for the labor-supply analysis. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 3: Summary statistics of estimated household shadow wages, shadow income, and market wages

	Round 1 (2010–11)			Round 2 (2015)			Round 3 (2018–19)		
Variable	N	Mean	SD	N	Mean	SD	N	Mean	SD
Panel A: Cobb–Douglas production function									
Female shadow wage (usd/hr) ($\widehat{w}^{*,f}$)	1464	0.025	0.170	1464	0.031	0.270	1464	0.036	0.226
Male shadow wage (usd/hr) ($\widehat{w}^{*,m}$)	1464	0.508	0.412	1464	0.726	0.594	1464	0.968	0.986
Shadow income per capita (usd/year) ($\widehat{y}^{*}$)	1464	338.074	566.626	1464	408.446	444.560	1464	521.562	531.558
Panel B: Translog production function									
Female shadow wage (usd/hr) ($\widehat{w}^{*,f}$)	1451	0.013	0.018	1453	0.017	0.026	1450	0.024	0.023
Male shadow wage (usd/hr) ($\widehat{w}^{*,m}$)	1191	0.794	0.627	1108	1.098	0.842	1252	1.651	1.524
Shadow income per capita (usd/year) ($\widehat{y}^{*}$)	1464	337.947	566.681	1464	408.221	444.532	1464	521.435	531.722
Panel C: Observed agricultural market wages									
Female market wage (usd/hr) (w^{f})				141	0.271	0.090	116	0.363	0.096
Male market wage (usd/hr) (w^{m})				254	0.420	0.092	256	0.568	0.145

Notes: The table reports descriptive statistics for the estimated household-specific shadow wages and shadow income recovered from the household production function. Female and male shadow wages are calculated as the marginal value products of family labor using Equation (5), while shadow income is constructed as the predicted value of poultry production net of variable production costs plus household income from non-poultry sources and addition of all exogenous income, as defined in Equation (6). The shadow income is presented on a per-capita basis. All monetary values are expressed in constant 2019 U.S. dollars.

Table 4: Jacoby (1993) test of shadow wage equality

	$\widehat{w}^{*,f}$		$\widehat{w}^{*,m}$	
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
Female agri. market wage (usd/hour)	0.010	-0.042		
	[0.076]	[0.153]		
Male agri. market wage (usd/hour)			-0.008	-0.267
			[0.224]	[0.568]
Constant	0.028**	0.042	0.730***	0.836***
	[0.014]	[0.037]	[0.096]	[0.237]
Year FE	x	x	x	x
Observations	1510	1510	2893	2893
Jacoby test: F-stat/χ^2	267.852	507.297	104.859	198.451
P-value	0.000	0.000	0.000	0.000

Notes: The table reports Jacoby (1993) separability test, which compares household-specific shadow wages with observed market wages. The dependent variables are the estimated female and male shadow wages. The test evaluates the joint null hypothesis that the intercept equals zero and the coefficient on the corresponding market wage equals one. Rejection of the null implies that observed market wages do not fully represent the opportunity cost of family labor, providing evidence against separability. Standard errors are clustered at the village level. Columns (2) and (4) report IV estimates.

Table 5: Le (2010) generalized test of separability

	$\ln(PQ/L)$	
	OLS	IV
	(1)	(2)
$\ln W^m$	0.248	0.310
	[0.191]	[0.588]
$\ln W^f$	-0.058	0.112
	[0.135]	[0.370]
Log of family female labor (hours/year)	-0.437***	-0.440***
	[0.054]	[0.057]
Log of family male labor (hours/year)	-0.093***	-0.108***
	[0.021]	[0.023]
Log of cost for feeding, medicine and others	0.067***	0.086***
	[0.025]	[0.029]
Log of initial livestock	0.276***	0.267***
	[0.026]	[0.030]
Log of productive asset	0.037***	0.031**
	[0.013]	[0.014]
Distance to nearest town (km)	0.000	0.002
	[0.002]	[0.002]
Crop and Livestock shcok	0.003	0.069
	[0.088]	[0.099]
Number of adult male	0.011	0.000
	[0.035]	[0.038]
Number of adult female	0.009	0.017
	[0.031]	[0.035]
Dependency ratio	0.021	0.032
	[0.045]	[0.051]
Average year of eduction for adults	0.018**	0.017**
	[0.008]	[0.009]
Year FE	x	x
Observations	1501	1236
Le generalized test:		
F-stat/χ^2	29.049	29.502
P-value	0.000	0.000

Notes: The dependent variable is the log of the value of poultry output per unit of labor input, $\ln(PQ/L)$. Column (1) reports ordinary least squares (OLS) estimates, while column (2) reports instrumental variables (IV) estimates, which constitute the preferred specification. Both models include year fixed effects and production controls. Standard errors clustered at the village level are reported in brackets. The Le generalized test evaluates the joint null hypothesis that the coefficients on the male and female market wages equal one and that household demographic characteristics contain no additional information for production decisions beyond that captured by market wages, observed labor inputs, and other production inputs. Rejection of the null indicates that production and household labor allocation decisions are jointly determined, providing evidence against separability. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6: First-stage relationships between excluded instruments and endogenous variables

	Female		**Male**	
	$\ln \widehat{w}^{*,f}$	$\ln \widehat{y}^{*}$	$\ln \widehat{w}^{*,m}$	$\ln \widehat{y}^{*}$
	(1)	(2)	(3)	(4)
$\ln P^{chi}$	-0.090	-0.015	1.262**	0.196
	[0.422]	[0.377]	[0.619]	[0.353]
$\ln P^{egg}$	0.829	0.318	0.935	0.244
	[0.627]	[0.642]	[0.770]	[0.595]
Log of household assets	0.043	0.147***	0.216***	0.107**
	[0.050]	[0.043]	[0.072]	[0.045]
Log of asset $\times \ln P^{chi}$	0.002	0.082	-0.255**	0.060
	[0.073]	[0.063]	[0.111]	[0.065]
Log of asset $\times \ln P^{egg}$	-0.034	0.022	0.014	0.040
	[0.098]	[0.100]	[0.117]	[0.100]
Log of initial poultry capital	0.172***	0.116**	0.252***	0.131**
	[0.049]	[0.047]	[0.070]	[0.053]
Log of initial poultry capital $\times \ln P^{chi}$	0.055	-0.120*	0.032	-0.151*
	[0.074]	[0.072]	[0.109]	[0.082]
Log of initial poultry capital $\times \ln P^{egg}$	-0.096	-0.057	-0.221*	-0.044
	[0.117]	[0.100]	[0.124]	[0.104]
Age	0.000	0.000	-0.002	-0.003*
	[0.001]	[0.001]	[0.002]	[0.001]
Married (=1)	0.004	0.153***	0.103	0.049
	[0.030]	[0.033]	[0.079]	[0.036]
literate	0.030	-0.006	-0.005	0.030
	[0.041]	[0.039]	[0.073]	[0.040]
Number of children	0.000	-0.155***	-0.010	-0.154***
	[0.016]	[0.017]	[0.024]	[0.015]
Number of elderly members	-0.046	-0.088*	-0.017	-0.143***
	[0.041]	[0.045]	[0.075]	[0.045]
Number of adults in the household	0.019	-0.010	0.025	-0.016
	[0.020]	[0.014]	[0.030]	[0.013]
Age of household head	0.001	-0.002	-0.002	0.000
	[0.002]	[0.002]	[0.003]	[0.002]
Household head is male	0.033	0.075	0.033	0.017
	[0.051]	[0.058]	[0.130]	[0.074]
Household head is literate	-0.006	0.016	0.090	0.006
	[0.043]	[0.043]	[0.091]	[0.049]
Landholding per capita (decimals)	0.002***	0.003***	0.002	0.004***
	[0.001]	[0.000]	[0.001]	[0.000]
Household has a loan	-0.013	0.056	-0.089	0.026
	[0.037]	[0.041]	[0.066]	[0.038]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	6396	6396	5283	5283
Partial R^2	0.068	0.061	0.045	0.041
Sanderson–Windmeijer conditional F	35.374	21.588	13.859	14.014

Notes: This table reports the first-stage regressions for the IV labor-supply models. Columns (1) and (2) present the first-stage equations for the female sample, and columns (3) and (4) present the corresponding equations for the male sample. The endogenous variables are household-specific shadow wages and household-specific shadow income. The excluded instruments are interactions between the logarithm of village-level poultry prices (chicken and eggs) and predetermined household productive assets, including initial poultry capital and household assets per capita. All specifications include the full set of individual, household, and community controls and survey-round fixed effects. Standard errors clustered at the village level are reported in brackets. Partial R^2 and Sanderson–Windmeijer (SW) conditional first-stage F statistics are reported for each endogenous variable. Kleibergen–Paap and Hansen's J statistics are reported with the corresponding second-stage estimates in Table 7. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

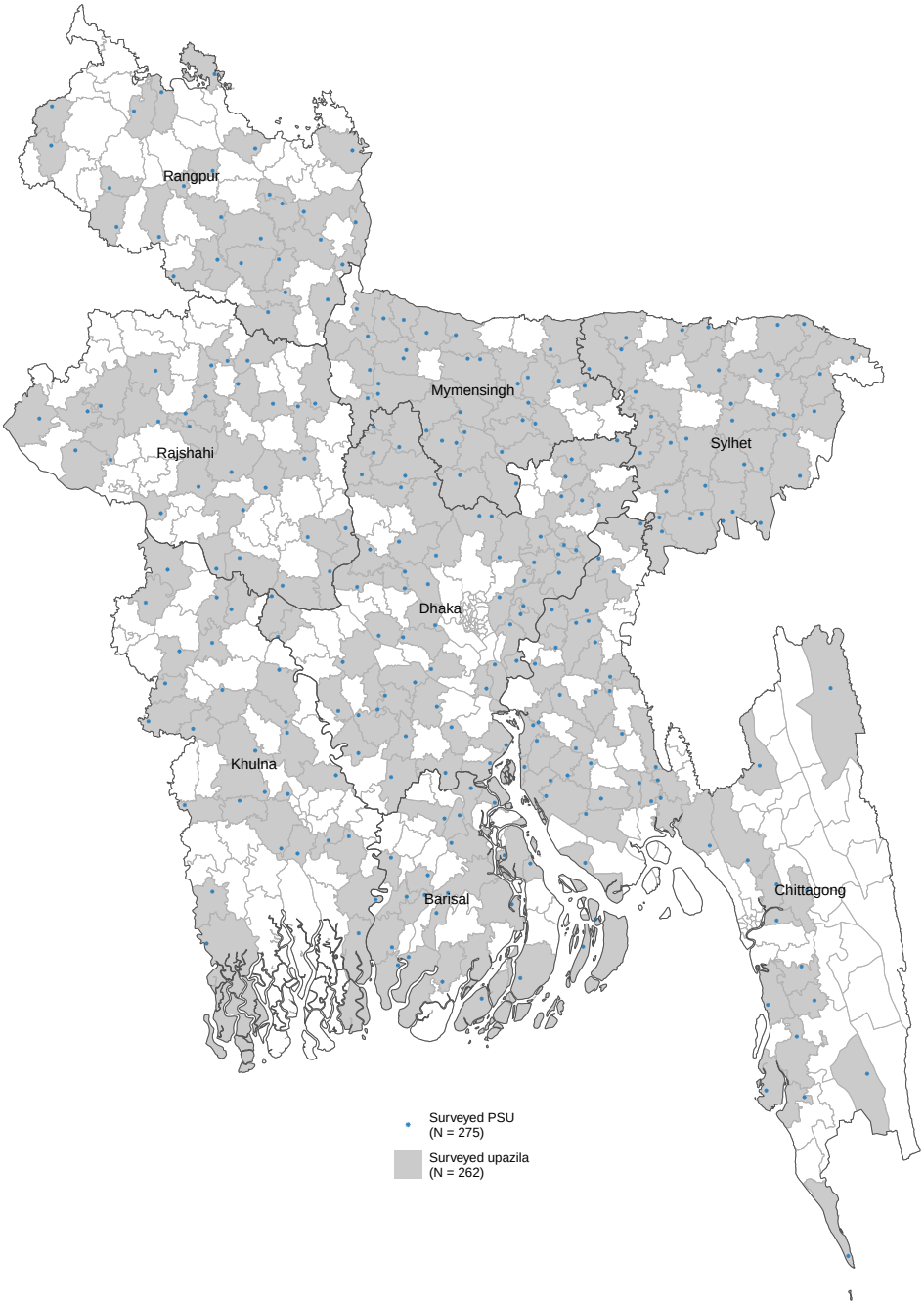
Table 7: IV estimated of the effects of household-specific shadow wages and shadow income on weekly labor supply

	Poultry labor supply		**Overall labor supply**	
	Participation	**Weekly hours**	**Participation**	**Weekly hours**
	(1)	(2)	(3)	(4)
Panel A: Female				
$\ln \widehat{w}^{*,f}$	0.075***	0.947***	0.016	-0.202
	[0.023]	[0.244]	[0.021]	[0.742]
$\ln \widehat{y}^{*}$	-0.061**	-0.297	-0.067***	-1.732*
	[0.026]	[0.247]	[0.025]	[0.947]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	6396	6396	6396	6396
Outcome mean	0.720	3.976	0.807	10.535
Instrument diagnostics:				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	20.378	20.378	20.378	20.378
Hansen J test: p-value	0.660	0.448	0.166	0.008
Panel B: Male				
$\ln \widehat{w}^{*,m}$	0.004	0.068	-0.008	-1.713
	[0.007]	[0.053]	[0.016]	[1.333]
$\ln \widehat{y}^{*}$	-0.015	0.034	-0.078**	4.875*
	[0.011]	[0.073]	[0.032]	[2.583]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	5283	5283	5283	5283
Outcome mean	0.014	0.061	0.863	38.421
Instrument diagnostics:				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	8.659	8.659	8.659	8.659
Hansen J test: p-value	0.954	0.310	0.827	0.276

Notes: The table reports IV estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the balanced panel of poultry-producing households observed in all three BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The endogenous regressors are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita. These endogenous regressors are instrumented using interactions between the logarithms of village-level chicken and egg prices and predetermined household characteristics, namely poultry capital at the beginning of the production year and household assets per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.2, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. The Kleibergen–Paap LM statistic tests the null hypothesis that the model is underidentified, the Kleibergen–Paap rk Wald F statistic assesses the strength of the excluded instruments under clustered standard errors, and Hansen's J statistic tests the validity of the overidentifying restrictions. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

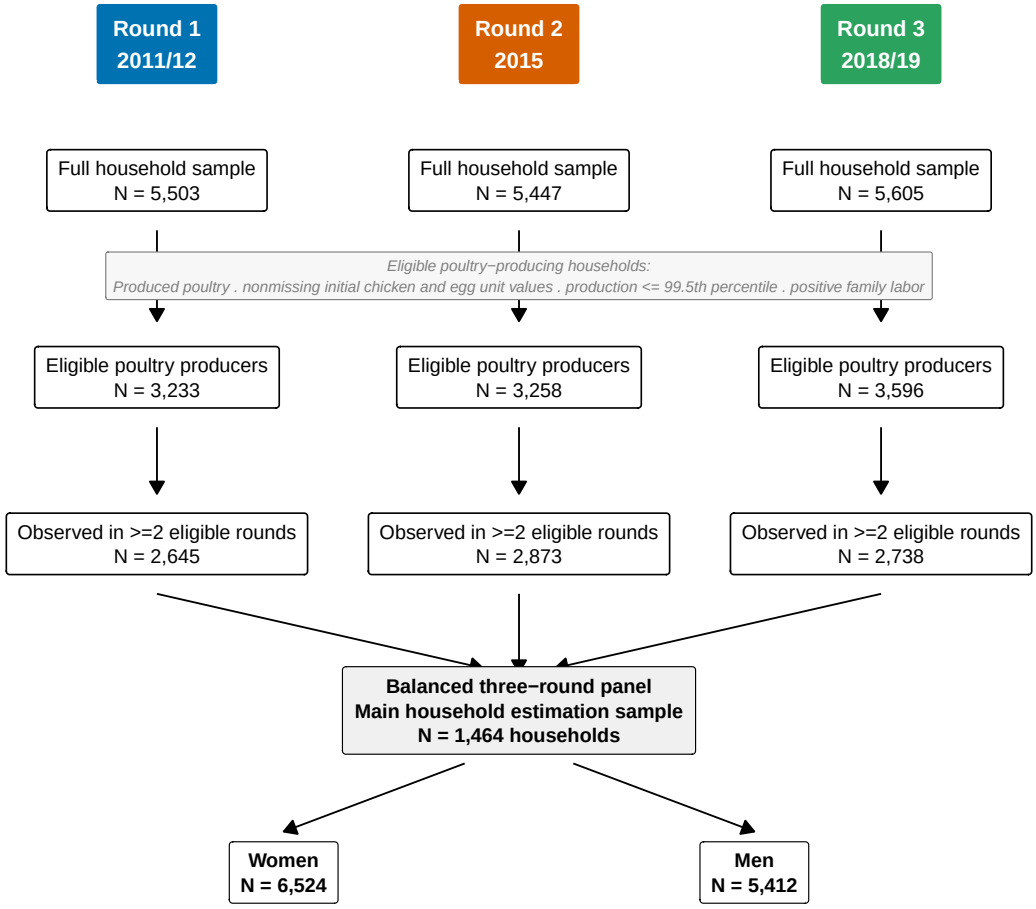
Appendix A. Figures and Tables

Figure A.1: Survey area of Bangladesh Integrated Household Survey



Notes: The production-function estimations use household observations. Labor supply estimations use individual observations for working-age household members (15–64 years). The primary estimation sample consists of the balanced panel of poultry-producing households observed in all three survey rounds. Robustness analyses additionally use poultry-producing households observed in at least two survey rounds.

Figure A.2: Sample selection for the empirical analysis



Notes: Household observations are used to estimate the production function. Individual observations include working-age household members (15–64 years) used in the labor supply analysis. The primary estimation sample consists of the balanced panel of poultry-producing households observed in all three survey rounds. Robustness analyses additionally use poultry-producing households observed in at least two survey rounds. Observations from two villages are excluded from both producer samples because village-level poultry prices could not be reliably constructed owing to insufficient household purchase observations.

Table A.1: Comparison of the main estimation sample and excluded poultry-producing households

	Round 1 (2010–11)			Round 2 (2015)			Round 3 (2018–19)		
Variable	Excluded	Estimation	Norm. Diff.	Excluded	Estimation	Norm. Diff.	Excluded	Estimation	Norm. Diff.
Panel A: Female characteristics									
Outcomes									
Poultry participation	0.644	0.672	0.059	0.698	0.722	0.053	0.766	0.760	-0.013
Poultry weekly hours	3.299	3.660	0.093	3.722	4.105	0.090	4.310	4.124	-0.032
Overall participation	0.750	0.781	0.073	0.790	0.802	0.031	0.848	0.835	-0.036
Overall weekly hours	9.710	10.438	0.056	9.961	10.428	0.038	11.491	10.703	-0.059
Characteristics									
Age	33.899	33.675	-0.018	34.692	34.769	0.006	34.840	35.480	0.048
Married	0.777	0.782	0.012	0.770	0.761	-0.019	0.753	0.768	0.036
Years of education	3.978	4.295	0.082	4.449	4.787	0.085	4.936	5.164	0.055
Panel B: Male characteristics									
Outcomes									
Poultry participation	0.008	0.008	0.001	0.006	0.007	0.013	0.038	0.028	-0.057
Poultry weekly hours	0.063	0.023	-0.053	0.025	0.077	0.057	0.111	0.082	-0.039
Overall participation	0.875	0.883	0.024	0.858	0.846	-0.032	0.872	0.861	-0.033
Overall weekly hours	40.326	41.372	0.042	38.978	37.364	-0.061	37.417	36.725	-0.026
Characteristics									
Age	34.858	35.696	0.061	35.056	35.720	0.046	35.730	35.156	-0.040
Married	0.711	0.732	0.047	0.696	0.683	-0.028	0.688	0.655	-0.070
Years of education	4.365	4.701	0.079	4.792	5.130	0.078	5.130	5.493	0.082
Panel C: Household Characteristics									
Household size	4.246	4.361	0.073	4.434	4.482	0.027	4.357	4.539	0.104
Number of children	1.554	1.643	0.077	1.529	1.546	0.015	1.486	1.457	-0.025
Number of adult males	1.091	1.137	0.060	1.168	1.240	0.084	1.141	1.261	0.140
Number of adult females	1.356	1.376	0.032	1.479	1.489	0.013	1.450	1.546	0.126
Number of elderly members	0.245	0.204	-0.088	0.258	0.206	-0.110	0.280	0.275	-0.010
Average years of adult education	3.977	4.205	0.071	4.355	4.621	0.085	4.757	5.042	0.091
Household assets	429.372	466.771	0.044	551.598	660.901	0.121	739.964	910.955	0.155
Homestead land (decimals)	10.809	11.521	0.054	10.003	11.568	0.126	10.000	12.017	0.156
Agricultural land (decimals)	82.043	99.437	0.114	78.360	105.107	0.162	69.643	101.575	0.256
Distance to main road (km)	2.055	1.976	-0.021	2.043	1.784	-0.064	2.248	2.206	-0.011
Distance to local market (km)	0.636	0.653	0.025	0.480	0.519	0.065	0.771	0.635	-0.042
Distance to district market (km)	1.821	1.821	0.000	1.726	1.727	0.001	1.919	1.916	-0.002
Experienced crop/livestock shock	0.118	0.130	0.039	0.070	0.075	0.019	0.086	0.125	0.126
Number of female	1601	2015		2084	2180		1847	2263	
Number of male	1289	1665		1646	1816		1454	1846	
Number of households	1181	1464		1409	1464		1274	1464	

Notes: The estimation sample (Est. sample) consists of poultry-producing households observed in all three survey rounds (balanced panel). The excluded sample (Exc. sample) consists of poultry-producing households observed in exactly two survey rounds. Individual characteristics are reported for working-age household members (15–64 years), while household characteristics are reported at the household level. Normalized differences are calculated as the difference in group means divided by the square root of the average of the two group variances. Absolute normalized differences below 0.10 generally indicate negligible differences between the two groups.

Table A.2: Definitions and measurement of variables used in the empirical analysis

Variable	Definition	Unit or construction
Panel A: Labor outcomes		
Poultry participation	Indicator equal to one if the individual worked in poultry production during the previous seven days, and zero otherwise.	Indicator (0/1)
Poultry weekly hours	Total hours devoted to poultry production during the previous seven days.	Hours per week
Overall participation	Indicator equal to one if the individual participated in any income-generating activity during the previous seven days, including salaried employment, wage work, self-employment, apprenticeship, or work in a family enterprise.	Indicator (0/1)
Overall weekly hours	Total hours devoted to all income-generating activities during the previous seven days.	Hours per week
Panel B: Individual characteristics		
Age	Age of the individual at the time of the survey.	Years
Married	Indicator equal to one if the individual is currently married, and zero otherwise.	Indicator (0/1)
Years of education	Number of completed years of formal schooling.	Years
Panel C: Household characteristics		
Household size	Total number of individuals residing in the household.	Number of persons
Number of children	Number of household members younger than 15 years.	Number of persons
Number of adult males	Number of male household members aged 15–64 years.	Number of persons
Number of adult females	Number of female household members aged 15–64 years.	Number of persons
Number of elderly members	Number of household members aged 65 years or older.	Number of persons
Average years of adult education	Average completed years of schooling among working-age household members aged 15–64 years.	Years
Household assets	Total value of household productive and other included assets before the annual poultry production.	Constant 2019 USD
Household income excluding poultry	Annual household income from sources other than poultry production.	Constant 2019 USD per year
Homestead land	Total area of land used for the household dwelling and surrounding homestead.	Decimals
Agricultural land	Total area of agricultural land operated or owned by the household, according to the measure used in the empirical analysis.	Decimals
Distance to main road	Distance from the household or village to the nearest main road.	Kilometers
Distance to local market	Distance from the household or village to the nearest local market.	Kilometers
Distance to district market	Distance from the household or village to the relevant district-level market.	Kilometers
Access to electricity	Indicator equal to one if the household has access to electricity, and zero otherwise.	Indicator (0/1)
Experienced crop or livestock shock	Indicator equal to one if the household experienced a crop- or livestock-related production shock during the survey reference period.	Indicator (0/1)
Household has a loan	Indicator equal to one if the household reported an outstanding loan or access to credit during the survey period.	Indicator (0/1)
Panel D: Poultry production variables		
Value of poultry production	Total annual value of household poultry output, including chickens and eggs produced during the survey year.	Constant 2019 USD per year
Initial poultry assets	Value of poultry stock held at the beginning of the production period.	Constant 2019 USD
Feed, medicine, and other production costs	Annual expenditures on poultry feed, veterinary medicine, and other variable poultry-production inputs.	Constant 2019 USD per year
Female family labor	Total annual hours supplied by female household members to poultry production.	Hours per year
Male family labor	Total annual hours supplied by male household members to poultry production.	Hours per year
Panel E: Local poultry prices		
Chicken price	Leave-one-out village-level median chicken unit value constructed from household-reported values and quantities of chicken stock at the beginning of the production period.	Constant 2019 USD per piece
Egg price	Leave-one-out village-level median egg unit value constructed from household-reported values and quantities of egg-producing stock at the beginning of the production period.	Constant 2019 USD per dozen
Panel F: Estimated mechanism variables		
Female shadow wage	Marginal value product of female family labor recovered from the estimated household poultry-production function.	Constant 2019 USD per hour
Male shadow wage	Marginal value product of male family labor recovered from the estimated household poultry-production function.	Constant 2019 USD per hour
Household-specific shadow income PC	Predicted poultry output value net of variable input costs and imputed family-labor costs, plus household income from external sources; divided by household size where the per-capita measure is used.	Constant 2019 USD per capita

Table A.3: Translog production function estimation for household poultry production

	Log of poultry output value		
	OLS	**Fixed Effect**	Random Effect
	(1)	(2)	(3)
Log of family female labor (hours/year)	1.386	-0.122	1.301
	[4.495]	[5.576]	[4.586]
Log of family male labor (hours/year)	1.706	1.741	1.702
	[3.757]	[4.281]	[3.753]
Log of feed, medicine, and other costs (2019 USD)	-2.355	-1.799	-2.354
	[4.834]	[5.351]	[4.927]
Log of initial poultry assets (2019 USD)	-8.927**	-8.635**	-8.931**
	[3.632]	[3.981]	[3.580]
Log of productive assets (2019 USD)	-0.593	-3.308	-0.712
	[1.796]	[2.650]	[1.807]
0.5 × [Log family female labor]^2	0.439	0.394	0.444
	[0.831]	[1.009]	[0.847]
0.5 × [Log family male labor]^2	0.556	0.743	0.565
	[0.815]	[1.041]	[0.880]
0.5 × [Log feed, medicine, and other costs]^2	4.203***	4.939***	4.227***
	[1.427]	[1.456]	[1.536]
0.5 × [Log initial poultry assets]^2	9.220***	7.566***	9.175***
	[0.994]	[1.271]	[1.136]
0.5 × [Log productive assets]^2	-0.693***	-0.972**	-0.698***
	[0.255]	[0.479]	[0.256]
Log family female labor × Log family male labor	-0.062	-0.084	-0.062
	[0.445]	[0.538]	[0.480]
Log family female labor × Log costs	0.275	0.206	0.276
	[0.668]	[0.763]	[0.693]
Log family female labor × Log initial poultry assets	-0.601	-0.470	-0.598
	[0.634]	[0.718]	[0.632]
Log family female labor × Log productive assets	0.220	0.616	0.236
	[0.293]	[0.461]	[0.290]
Log family male labor × Log feed, medicine, and other costs	-0.507	-0.910*	-0.523
	[0.478]	[0.491]	[0.495]
Log family male labor × Log initial poultry assets	-0.578	-0.581	-0.579
	[0.651]	[0.589]	[0.631]
Log family male labor × Log productive assets	0.164	0.193	0.165
	[0.204]	[0.255]	[0.204]
Log costs × Log initial assets	-0.769	-1.134	-0.782
	[0.668]	[0.730]	[0.701]
Log costs × Log productive assets	-0.298	0.046	-0.285
	[0.272]	[0.324]	[0.291]
Log initial poultry assets × Log productive assets	0.949***	1.177***	0.957***
	[0.270]	[0.267]	[0.236]
Household head is male (=1)	-1.072	-3.664	-1.101
	[1.483]	[2.849]	[1.559]
Age of the household head (years)	-0.021	0.021	-0.021
	[0.040]	[0.109]	[0.041]
Household head can read and write (=1)	1.360	4.186	1.386
	[1.103]	[2.969]	[1.194]
Crop and Livestock shock	-0.908	1.589	-0.807
	[1.644]	[2.073]	[1.696]
Distance to nearest market (km)	-0.083	-0.270	-0.088
	[0.274]	[0.332]	[0.308]
electricity	-0.514	-0.768	-0.526
	[1.177]	[2.298]	[1.352]
Constant	-2.138	11.420	-1.567
	[14.015]	[18.356]	[14.148]
Observations	4392	4392	4392
Adj. R^2	0.289	0.259	.
F(Fixed effect)/Wald(Random effect)		26.651	1103.510
F(Fixed effect)/Wald(Random effect): p-value		0.000	0.000
Hausman test χ^2		78.043	

Notes: The production function is estimated using Translog production function. The dependent variable is the log of the annual value of poultry output (2019 USD). Family male labor, family female labor, livestock capital, production expenditures, and productive assets are measured in logarithms. All specifications include survey-round fixed effects. Columns (2) and (3) report household fixed-effects (FE) and random-effects (RE) estimates, with standard errors clustered at the village level reported in brackets. The Hausman test evaluates the null hypothesis that the random-effects estimator is consistent; rejection of the null supports the use of the fixed-effects estimator. The Fixed Effect production-function estimates are subsequently used to recover household-specific shadow wages and shadow income for the labor-supply analysis. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.4: First stage results of the Jacoby Test of Shadow Wage Equality

	$\ln \widehat{w}^{*,f}$	$\ln \widehat{w}^{*,m}$
	(1)	(2)
Male non-agricultural wage	0.174*	0.324***
	[0.105]	[0.124]
Distance to nearest local market (km)	0.005	0.005
	[0.004]	[0.004]
Year FE	x	x
Observations	1510	2893
Partial R^2	0.083	0.171
Sanderson–Windmeijer conditional F	1.818	4.804

Notes: The table reports first-stage estimates for the instrumental-variable (IV) specifications of Jacoby's separability test. The dependent variable is the observed gender-specific agricultural market wage. The excluded instruments are village-level non-agricultural wages and distance to the nearest local market. All specifications include survey-round fixed effects. Standard errors are clustered at the village level.

Table A.5: First stage of the Le's generalized test of separability

	$\ln W^m$ (1)	$\ln W^f$ (2)
Male non-agricultural wage	0.467	0.158
	[0.287]	[0.193]
Female non-agricultural wage	0.620**	2.096***
	[0.276]	[0.289]
Distance to nearest local market (km)	0.017	0.012
	[0.017]	[0.009]
Log of family female labor (hours/year)	0.022*	0.033***
	[0.012]	[0.012]
Log of family male labor (hours/year)	0.007	0.007
	[0.006]	[0.006]
Log of cost for feeding, medicine and others	-0.020*	0.004
	[0.010]	[0.011]
Log of initial livestock	0.016***	0.003
	[0.006]	[0.007]
Log of productive asset	0.000	-0.005
	[0.004]	[0.006]
Distance to nearest town (km)	-0.001	-0.001
	[0.001]	[0.001]
Crop and Livestock shcok	-0.025	-0.012
	[0.018]	[0.022]
Number of adult male	0.009	0.025**
	[0.009]	[0.011]
Number of adult female	0.017**	0.026**
	[0.008]	[0.012]
Dependency ratio	0.025**	0.036**
	[0.011]	[0.014]
Average year of eduction for adults	-0.000	-0.000
	[0.002]	[0.003]
Controls	x	x
Year FE	x	x
Observations	1236	1236
Partial R^2	0.213	0.326
Sanderson–Windmeijer conditional F	1.971	6.307

Notes: The table reports first-stage estimates for the instrumental-variable (IV) specification of Le's generalized separability test. The dependent variables are the logarithms of male and female agricultural market wages. The excluded instruments are village-level non-agricultural wages and distance to the nearest local market. All specifications include household demographic characteristics, production controls, and survey-round fixed effects. Standard errors clustered at the village level are reported in brackets. First-stage F-statistics assess the relevance of the excluded instruments. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.6: OLS estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply

	Poultry labor supply		**Overall labor supply**	
	Participation	**Weekly hours**	**Participation**	**Weekly hours**
	(1)	(2)	(3)	(4)
Panel A: Female				
$\ln \widehat{w}^{*,f}$	-0.011*	-0.294***	-0.016***	-0.485***
	[0.006]	[0.067]	[0.005]	[0.183]
$\ln \widehat{y}^{*}$	0.004	-0.051	0.007	0.341*
	[0.005]	[0.048]	[0.004]	[0.178]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	6396	6396	6396	6396
Outcome mean	0.720	3.976	0.807	10.535
Panel B: Male				
$\ln \widehat{w}^{*,m}$	-0.013***	-0.057***	-0.006*	0.155
	[0.002]	[0.018]	[0.003]	[0.240]
$\ln \widehat{y}^{*}$	-0.001	0.015	0.035***	5.695***
	[0.003]	[0.014]	[0.005]	[0.430]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	5283	5283	5283	5283
Outcome mean	0.014	0.061	0.863	38.421

Notes: The table reports ordinary least squares (OLS) estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the balanced panel of poultry-producing households observed in all three BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The regressors of interest are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.2, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.7: IV labor-supply estimates using shadow wages and shadow income recovered from the translog production function

	Poultry labor supply		Overall labor supply	
	Participation	**Weekly hours**	**Participation**	**Weekly hours**
	(1)	(2)	(3)	(4)
Panel A: Female				
$\ln \widehat{w}^{*,f}$	-0.213***	-2.232**	-0.093	1.286
	[0.076]	[0.929]	[0.072]	[3.110]
$\ln \widehat{y}^{*}$	0.042	0.876*	-0.027	-2.269*
	[0.035]	[0.464]	[0.032]	[1.301]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	6341	6341	6341	6341
Outcome mean	0.721	3.979	0.808	10.550
Instrument diagnostics:				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	4.109	4.109	4.109	4.109
Hansen J test: p-value	0.715	0.291	0.190	0.012
Panel B: Male				
$\ln \widehat{w}^{*,m}$	0.006	0.085	-0.028	10.383**
	[0.022]	[0.081]	[0.051]	[4.929]
$\ln \widehat{y}^{*}$	-0.011	-0.019	-0.070**	0.665
	[0.014]	[0.065]	[0.035]	[3.568]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	4231	4231	4231	4231
Outcome mean	0.014	0.049	0.867	38.443
Instrument diagnostics:				
Kleibergen-Paap LM test: p-value	0.031	0.031	0.031	0.031
Kleibergen-Paap rk Wald F statistic	2.087	2.087	2.087	2.087
Hansen J test: p-value	0.963	0.909	0.925	0.711

Notes: The table reports IV estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the balanced panel of poultry-producing households observed in all three BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The endogenous regressors are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita, where the shadow variables are recovered from the translog production function. These endogenous regressors are instrumented using interactions between the logarithms of village-level chicken and egg prices and predetermined household characteristics, namely poultry capital at the beginning of the production year and household assets per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.2, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. The Kleibergen–Paap LM statistic tests the null hypothesis that the model is underidentified, the Kleibergen–Paap rk Wald F statistic assesses the strength of the excluded instruments under clustered standard errors, and Hansen's J statistic tests the validity of the overidentifying restrictions. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.8: Robustness check: IV labor-supply estimates using the broader sample of poultry-producing households observed in at least two survey rounds

	Poultry labor supply		**Overall labor supply**	
	Participation	**Weekly hours**	**Participation**	**Weekly hours**
	(1)	(2)	(3)	(4)
Panel A: Female				
$\ln \widehat{w}^{*,f}$	0.091***	0.990***	0.061***	1.548**
	[0.023]	[0.234]	[0.021]	[0.727]
$\ln \widehat{y}^{*}$	-0.081***	-0.301	-0.102***	-2.741***
	[0.023]	[0.227]	[0.022]	[0.883]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	11867	11867	11867	11867
Outcome mean	0.713	3.886	0.803	10.470
Instrument diagnostics:				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	24.012	24.012	24.012	24.012
Hansen J test: p-value	0.373	0.105	0.040	0.001
Panel B: Male				
$\ln \widehat{w}^{*,m}$	-0.001	0.018	-0.001	-0.833
	[0.006]	[0.035]	[0.011]	[0.853]
$\ln \widehat{y}^{*}$	0.003	0.097*	-0.044*	5.126**
	[0.008]	[0.054]	[0.024]	[1.981]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	9630	9630	9630	9630
Outcome mean	0.016	0.062	0.865	38.606
Instrument diagnostics:				
Kleibergen-Paap LM test: p-value	0.000	0.000	0.000	0.000
Kleibergen-Paap rk Wald F statistic	16.723	16.723	16.723	16.723
Hansen J test: p-value	0.473	0.402	0.490	0.683

Notes: The table reports IV estimates of the effects of household-specific shadow wages and shadow income on weekly labor supply for working-age individuals (15–64 years) from the broader sample of poultry-producing households observed in at least two BIHS survey rounds. Panel A reports estimates for women and Panel B reports estimates for men. The endogenous regressors are the logarithms of the household-specific female and male shadow wages and the inverse hyperbolic sine of household-specific shadow income per capita. These endogenous regressors are instrumented using interactions between the logarithms of village-level chicken and egg prices and predetermined household characteristics, namely poultry capital at the beginning of the production year and household assets per capita. All specifications include the full set of individual, household, and village control variables described in Appendix Table A.2, together with survey-round fixed effects. Standard errors clustered at the village level are reported in parentheses. The Kleibergen–Paap LM statistic tests the null hypothesis that the model is underidentified, the Kleibergen–Paap rk Wald F statistic assesses the strength of the excluded instruments under clustered standard errors, and Hansen's J statistic tests the validity of the overidentifying restrictions. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.9: Reduced form estimate of the IV

	Poultry labor supply		Overall labor supply	
	Participation	**Weekly hours**	**Participation**	**Weekly hours**
	(1)	(2)	(3)	(4)
Panel A: Female				
$\ln P^{chi}$	-0.218	0.202	0.001	5.022
	[0.141]	[1.459]	[0.117]	[3.642]
$\ln P^{egg}$	0.301	3.080	0.038	-0.805
	[0.198]	[1.969]	[0.151]	[4.741]
Log of household assets	-0.023	0.142	-0.008	0.623
	[0.016]	[0.158]	[0.013]	[0.434]
Log of asset $\times \ln P^{chi}$	0.024	-0.184	-0.001	-1.083*
	[0.022]	[0.223]	[0.018]	[0.573]
Log of asset $\times \ln P^{egg}$	-0.033	-0.197	-0.022	-0.568
	[0.026]	[0.268]	[0.021]	[0.697]
Log of initial poultry capital	0.008	0.077	0.017	0.279
	[0.012]	[0.136]	[0.012]	[0.426]
Log of initial poultry capital $\times \ln P^{chi}$	0.014	0.239	-0.019	-0.430
	[0.017]	[0.203]	[0.017]	[0.623]
Log of initial poultry capital $\times \ln P^{egg}$	-0.028	-0.570*	0.002	0.125
	[0.026]	[0.300]	[0.023]	[0.913]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	6396	6396	6396	6396
Outcome mean	0.720	3.976	0.807	10.535
Joint sign. F-stat	2.082	3.284	2.412	3.406
Joint sign. p-value	0.038	0.001	0.016	0.001
Panel B: Male				
$\ln P^{chi}$	0.027	0.744	-0.032	4.654
	[0.042]	[0.497]	[0.096]	[8.762]
$\ln P^{egg}$	-0.015	-0.406	-0.051	3.072
	[0.066]	[0.562]	[0.135]	[12.835]
Log of household assets	0.000	0.077	-0.014	1.302
	[0.005]	[0.055]	[0.011]	[0.979]
Log of asset $\times \ln P^{chi}$	-0.003	-0.109	0.002	-1.139
	[0.006]	[0.077]	[0.016]	[1.416]
Log of asset $\times \ln P^{egg}$	-0.002	0.069	-0.004	1.779
	[0.008]	[0.083]	[0.021]	[1.849]
Log of initial poultry capital	0.002	0.037	-0.016	-0.612
	[0.005]	[0.044]	[0.012]	[0.964]
Log of initial poultry capital $\times \ln P^{chi}$	-0.003	-0.049	0.015	0.584
	[0.007]	[0.064]	[0.018]	[1.347]
Log of initial poultry capital $\times \ln P^{egg}$	0.005	0.072	0.023	-2.345
	[0.011]	[0.105]	[0.023]	[1.787]
Controls	x	x	x	x
Year FE	x	x	x	x
Observations	5283	5283	5283	5283
Outcome mean	0.014	0.061	0.863	38.421
Joint sign. F-stat	0.390	1.016	1.722	1.341
Joint sign. p-value	0.926	0.424	0.094	0.223

Notes: Standard errors clustered at the village level are reported in brackets. Partial R^2 and Sanderson–Windmeijer (SW) conditional first-stage F statistics are reported for each endogenous variable. Kleibergen–Paap and Hansen's J statistics are reported with the corresponding second-stage estimates in Table 7. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix B. Inverse Hyperbolic Sine Transformation

B.1 IHS Transformation

Several production inputs used to estimate the household poultry production function contain a substantial number of zero observations. In particular, many backyard poultry-producing households report no male family labor or no expenditures on feed, veterinary medicine, and other purchased inputs during the production year. Estimating the production function using natural logarithms would therefore require excluding these observations or adding arbitrary constants to zero-valued variables, both of which may introduce sample selection or measurement error.

To retain all production observations while preserving the nonlinear relationship between inputs and output, continuous production inputs are transformed using the inverse hyperbolic sine (IHS) transformation. For any non-negative variable x, the transformation is defined as

$$\sinh^{-1}(x) = \ln\left(x + \sqrt{x^2 + 1}\right) \tag{17}$$

The IHS transformation is defined for zero-valued observations and closely approximates the natural logarithm for sufficiently large values. Specifically,

$$\text{IHS}(x) \approx \ln(2x), \qquad \text{as } x \to \infty. \tag{18}$$

Consequently, the transformation retains observations with zero input use while maintaining an interpretation similar to a logarithmic specification over most of the support of the data. The IHS transformation has therefore become a standard approach for handling zero-valued variables in applied microeconomic research (Bellemare and Wichman, 2020).

In the empirical analysis, the IHS transformation is applied to all continuous production inputs that exhibit zero observations, including female family labor, male family labor, expenditures on feed, veterinary medicine, and other variable poultry-production inputs, the value of the initial poultry stock, and productive household assets. The dependent variable, annual real value of poultry production, is strictly positive for all households included in the estimation sample and therefore enters the production function in natural logarithms.

Using the IHS transformation allows the production function to be estimated using the full balanced panel of poultry-producing households while avoiding the loss of observations associated with logarithmic transformations. The estimated production elasticities are subsequently used to recover household-specific shadow wages and shadow income, as described in Section 5.1. The theoretical agricultural household model presented in Online Appendix B is formulated in levels and is unaffected by this empirical transformation, which is introduced solely for estimation purposes.

B.2 Recovering Shadow Wages under the IHS Specification

As family labor enters the estimated production function using the inverse hyperbolic sine transformation, the corresponding marginal value product must account for the derivative of that transformation. For labor type

$g \in f, m$, the estimated production function implies Equation 11

$$\ln \widehat{V}_{ht} = \cdots + \hat{\beta}_g \mathbf{IHS}\left(L_{ht}^{g}\right) + \cdots . \tag{19}$$

Applying the chain rule gives

$$\frac{\partial \widehat{V}_{ht}}{\partial L_{ht}^{g}} = \frac{\partial \widehat{V}_{ht}}{\partial \mathbf{IHS}(L_{ht}^{g})} \frac{\partial \mathbf{IHS}(L_{ht}^{g})}{\partial L_{ht}^{g}}. \tag{20}$$

Since,

$$\frac{\partial \mathbf{IHS}(x)}{\partial x} = \frac{1}{\sqrt{x^2 + 1}}, \tag{21}$$

and

$$\frac{\partial \widehat{V}_{ht}}{\partial \mathbf{IHS}(L_{ht}^{g})} = \hat{\beta}_g \widehat{V}_{ht}, \tag{22}$$

the household-specific shadow wage is

$$\widehat{w}_{ht}^{*g} = \hat{\beta}_g \frac{\widehat{V}_{ht}}{\sqrt{\left(L_{ht}^{g}\right)^2 + 1}}. \tag{23}$$

This expression is well defined when reported family labor is zero, unlike the standard log–log Cobb–Douglas expression. For sufficiently large positive values of labor input,

$$\sqrt{\left(L_{ht}^{g}\right)^2 + 1} \approx L_{ht}^{g}, \tag{24}$$

so Equation (23) approaches the familiar marginal-value-product expression $\hat{\beta}_g \widehat{V}_{ht} / L_{ht}^{g}$.

Appendix C. Translog Production Function

The main analysis estimates household-specific shadow wages and shadow income using a Cobb–Douglas production function because it provides a parsimonious representation of household poultry production, facilitates the recovery of marginal value products of family labor, and is well suited to the relatively short household panel. As a robustness check, the production technology is also estimated using a translog specification, which relaxes the Cobb–Douglas assumption of constant output elasticities by allowing for second-order terms and interactions among production inputs.

The translog production function is specified as

$$\ln V_{ht} = \alpha + \sum_{k} \beta_k X_{k,ht} + \frac{1}{2} \sum_{k} \sum_{l} \gamma_{kl} X_{k,ht} X_{l,ht} + \lambda_t + \mu_h + \varepsilon_{ht}, \tag{25}$$

where V_{ht} denotes the annual real value of household poultry production, $X_{k,ht}$ includes the same set of transformed production inputs used in the main specification, λ_t denotes survey-round fixed effects, and μ_h represents household fixed effects. As in the baseline model, labor inputs are measured using the inverse hyperbolic sine (IHS) transformation to accommodate zero observations while preserving the panel structure.

Household-specific shadow wages are recovered as the marginal value products of female and male family labor by differentiating the estimated translog production function with respect to the corresponding labor input and evaluating the resulting expression at each household's observed input levels. Household-specific shadow income is then constructed using the predicted value of poultry production together with household production costs and non-poultry income, following the same procedure described in the main text.

Table A.3 reports the estimated translog production function. The estimated coefficients are broadly consistent with the baseline specification, indicating that the principal findings are not driven by the assumption of a Cobb–Douglas production technology.